BUSINESS OPERATING BLUEPRINT

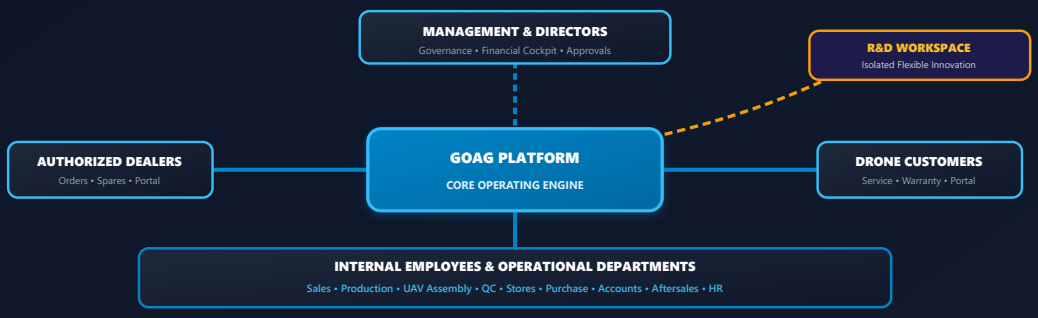
GOAG PLATFORM

BUSINESS FLOW & FUNCTIONAL ARCHITECTURE

Version 1.0 • Team Discussion & Approval Draft

OPERATING SCOPE & LIFECYCLE:
"How GOAG Services operates — from Lead → Sale → Payment → Work Order → Production → QC → Delivery → Aftersales → Finance → HR → Management"

GOAG PLATFORM CONNECTIVITY ECOSYSTEM



CORE BLUEPRINT OBJECTIVE
This document serves as the formal operational contract and functional blueprint for GOAG Services. It freezes business rules, employee responsibilities, document triggers, and cross-department workflows prior to software engineering.

Target Audience: Board, Directors & Employees
Status: Freeze Before Implementation
Verification: Full 118-Item Audit Included

© 2026 GOAG Services Pvt. Ltd. • Strictly Confidential

ARCHITECTURE GUIDE • TABLE OF CONTENTS

Operational Blueprint Index & Governance Rules

Comprehensive structural breakdown of all 46 functional sections across 8 core operational domains.

CONFIRMED REQUIREMENT
Explicitly mandated by source specification and business notes.

PROPOSED ADDITION
Architectural best-practice enhancement; pending leadership approval.

DECISION REQUIRED
Operational ambiguity or policy choice to be frozen before coding.

VERIFY BEFORE BUILD
Requires external statutory, Indian GST, or legal verification.

PART I: PLATFORM GOVERNANCE & HUMAN CAPITAL

Section 1: Executive Overview & Ecosystem Flow — Page 3
Section 2: Platform Organization & Topology — Page 4
Section 3: 23-Employee Master Directory — Page 5
Section 3 (Cont): Role-Based Access Control (RBAC) Matrix — Page 6

PART II: COMMERCIAL ENGINE & ORDER PIPELINE

Section 4: Master Data Architecture & GST Rule Engine — Page 7
Section 5: Macro Process: Complete Sales → Delivery Pipeline — Page 8
Section 6: Sales & Lead Management (Auto vs Custom Quotes) — Page 9
Section 7: Quotation → Payment → Work Order Gateway — Page 10
Section 8: Customer Account & Portal Provisioning — Page 11

PART III: DRONE MANUFACTURING & QUALITY ASSURANCE

Section 9: Production 6-Lane Swimlane & Time Tracking — Page 12
Section 10: Database-Driven 20-Page Checklist Architecture — Page 13
Sections 11 & 12: Test Flight Protocol & Quality Control (QC) — Page 14

PART IV: SUPPLY CHAIN, STORES & FINANCIAL CONTROLS

Section 13: Stores Inventory & Automated BOM Availability Check — Page 15
Section 14: Material Requests, Vendor Approvals & Purchase Orders — Page 16
Section 15: Purchase Tagging to Accounts & Expense Visibility — Page 17
Section 16: Dual-Path Petty Cash & Voucher Management — Page 18
Section 17: Cash vs Account Transaction Invoicing Distinction — Page 19
Section 18: Payment, Proforma & Delivery Trigger Logic — Page 20
Section 19: Dispatch, Logistics (Pickup/Shipping) & Ledger — Page 21

PART V: AFTERSALES SUPPORT & SERVICE OPERATIONS

Section 20: Inbound Support, Phone Resolution & Pattern Analysis — Page 22
Sections 21 & 22: Rotation & Branch Service Point Inward (OTP) — Page 23
Section 22 (Cont): Diagnostics, Service Checklist & OTP Handover — Page 24
Sections 23 & 24: Manager Override & External Service Centres — Page 25

PART VI: HR, EMPLOYEE SELF-SERVICE & EXECUTIVE COCKPIT

Section 25: HR Architecture & Attendance (Manual to Key-Card) — Page 26
Section 26: Dynamic HR Document Generation & Salary Sheet — Page 27
Sections 27 & 28: Employee Self-Service (ESS) & Dashboard UI — Page 28
Section 29: Management & Director Executive Cockpit — Page 29
Section 30: Geographic Delivery Map Architecture — Page 30

PART VII: EXTERNAL PORTALS & INNOVATION SANDBOX

Section 31: Dealer Portal Architecture & Orders — Page 31
Sections 32 & 33: Dealer Customer Mapping & Customer Portal UI — Page 32
Section 34: Isolated R&D Workspace & Product Handover Gate — Page 33

PART VIII: ENTERPRISE MASTER BLUEPRINT & TECHNICAL FOUNDATION

Section 35: Master "One-Page Picture" Operating Map — Page 34
Sections 36 & 37: System Architecture & Entity Relationships — Page 35
Sections 38 & 39: Document ID Inventory & Status Lifecycles — Page 36
Section 40: Automation vs Manual Responsibility Matrix — Page 37
Sections 41 & 42: Managerial Approvals & Notification Engine — Page 38
Section 43: Audit Logging & Historical Traceability — Page 39
Section 44: Open Decisions Required Before Development — Page 40
Section 45: Proposed Architectural Additions — Page 41
Section 46: Team Verification Checklist & Board Sign-off — Page 42

CONFIRMED SOURCE-OF-TRUTH MANDATE:
Every operational step, role assignment, and system trigger within this document reflects the exact operational logic provided in the GOAG source documents. Software development must adhere strictly to these approved functional requirements without generic ERP substitutions.

SECTION 1 • STRATEGIC ARCHITECTURE

Executive Overview & Operating Ecosystem

Defining the core mission, business rationale, user landscape, and interconnected operational flow of GOAG Services.

What is GOAG Platform? CONFIRMED

A unified, proprietary business operating system designed specifically for GOAG's high-tech drone manufacturing, aftermarket services, franchise dealer distribution, and cross-functional corporate operations. It replaces disparate manual spreadsheets with an authoritative single source of truth.

Why is GOAG Building It? CONFIRMED

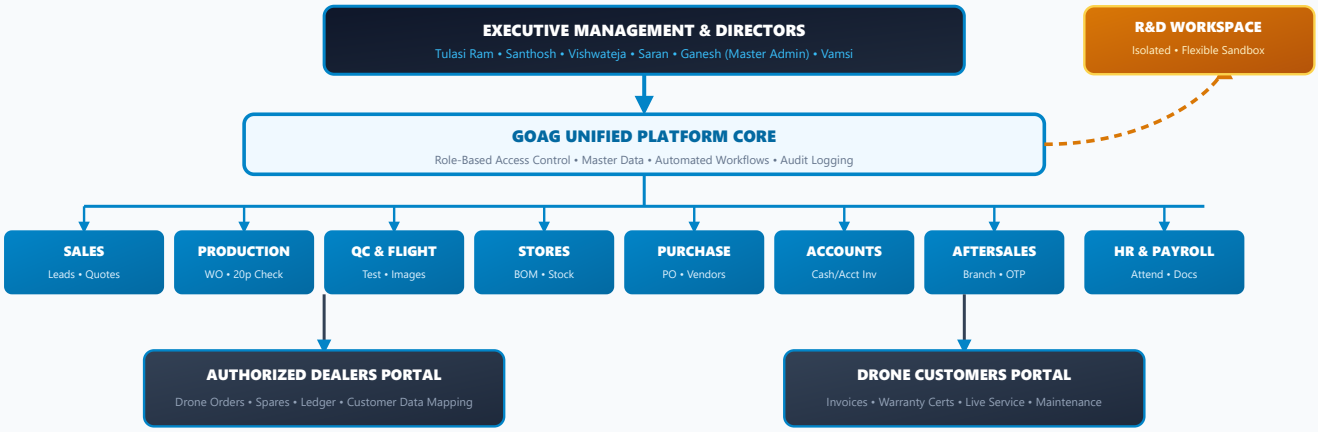
To enforce operational discipline across assembly and field teams, introduce structured timesheet tracking without imposing unvalidated deadlines, ensure 100% Bill-of-Materials (BOM) inventory availability before assembly, eliminate leakage in cash and credit invoicing, and provide real-time director-level financial governance.

Who Uses the Platform? CONFIRMED

Internal stakeholders across all 23 named employees (Directors, Sales Managers, UAV Assembly Technicians, Stores Supervisors, Accounts, QC Inspectors, HR) and external stakeholders (Authorized Drone Dealers and Commercial Drone Customers) via dedicated, role-tailored digital portals.

GOAG ENTERPRISE INFORMATION & OPERATIONAL FLOW ECOSYSTEM

CORE ECOSYSTEM ARCHITECTURE



Cross-Department Data Mesh: When Sales confirms a Work Order, the platform instantly engages Stores for BOM validation, alerts Production Leads for technician assignment, and provisions Customer Portal access without manual email relays.

Strict R&D Isolation: Unlike commercial assembly lines, the R&D Workspace (Lead: Voleti Veera Babu) remains unencumbered by rigid sales work orders, operating as an agile experimental sandbox with flexible data logging.

SECTION 2 • STRUCTURAL TOPOLOGY

Platform Organization & User Categorization

Defining user domains, departmental segmentation, external stakeholder boundaries, and access topologies.

1. EXECUTIVE MANAGEMENT TIER

CONFIRMED

Stakeholders: Cofounders & Directors (Yalamarthi Tulasi Ram, Bandaru Santhosh Kumar, V Vishwateja Kumar Reddy, Battula Saran Kumar, Ganesh Pentela, Vamsi Krishna K).
Operational Capabilities: Real-time financial cockpits (Cash In / Cash Out / Liabilities), sales pipeline tracking, production progress monitoring, marketing schedules, vendor management, high-level approvals for custom quotes, delayed payments, and capital purchases. Master ERP administration is centralized under Ganesh Pentela.

2. EMPLOYEE OPERATIONAL TIER

CONFIRMED

Scope: 9 Specialized Internal Operational Departments:
- *Sales:* Lead tracking, auto/custom quote generation, client delivery date setup.
- *Production:* Work order intake, technician assignment, timesheet tracking.
- *UAV Assembly:* Step-by-step completion of the 20-page drone assembly checklist.
- *Quality Control:* Comprehensive physical/functional audit, mandatory photo upload.
- *Stores & Inventory:* Live BOM checking, stock inward/outward, dispatch packaging.
- *Procurement:* Material shortage resolution, vendor orders, purchase approvals.
- *Accounts:* Cash vs Account transaction invoicing, petty cash, e-way bills.
- *Aftersales:* Phone troubleshooting, service point intake, warranty overrides.
- *HR:* Attendance tracking, salary sheets, document generation (offers, payslips).

3. AUTHORIZED DEALER TIER

CONFIRMED

Target Users: Franchise Drone Dealerships & Regional Distributors.
Capabilities: Dedicated restricted portal for placing bulk drone and spare parts orders, tracking order fulfillment, viewing ledger balances, logging dealer payments for accounts reconciliation, submitting warranty service requests directly to Vikas, registering end-customer sales data, and accessing authorized marketing schedules.

4. COMMERCIAL CUSTOMER TIER

CONFIRMED

Target Users: Commercial Drone Buyers, Agricultural Operators & Enterprises.
Capabilities: Streamlined portal providing permanent access to tax invoices, official warranty certificates uploaded by GOAG, real-time stage-by-stage status of drones currently in service, historical maintenance records, and automated scheduled maintenance alerts.

PLATFORM FUNCTIONAL BOUNDARY & ISOLATION ARCHITECTURE

User Domain	Primary Interface	Authorized Operational Scope	Data Isolation Level
Executive Management	Director Cockpit & Master Console	Full visibility across all finances, WIP, sales, vendors, and strategic approvals.	UNRESTRICTED
Internal Employees	Employee Self-Service (ESS) & Role Apps	Department-specific tasks, checklists, timesheets, and leave/payslip requests.	ROLE-RESTRICTED (No financial data in WO)
Authorized Dealers	Dealer Web Portal	Own orders, spare parts catalog, own ledger, end-customer registration.	TENANT-ISOLATED
Drone Customers	Customer Web/Mobile Portal	Own purchased aircraft, invoices, warranty docs, live service tickets.	CUSTOMER-ISOLATED
R&D Engineering	R&D Innovation Workspace	Flexible prototyping, test iterations, raw CAD/notes; exempt from rigid WO rules.	SEPARATE SANDBOX

CONFIRMED ARCHITECTURAL BOUNDARY MANDATE:
External dealers and customers have strictly walled access to only their own operational records. Production assembly personnel are strictly restricted from seeing financial data, purchase costs, or profit margins on work orders. Master ERP access and company-wide data editing is reserved solely for Ganesh Pentela.

SECTION 3.1 • HUMAN CAPITAL DIRECTORY

Authoritative 23-Employee Master Directory

Exhaustive personnel listing with official employee IDs, organizational designations, and core system responsibilities.

PRIMARY SOURCE OF TRUTH: All 23 employee profiles, official IDs (GOAG-A001 to GOAG-A006 and GOAG-E007 to GOAG-E023), and mandated system roles match the authoritative employee specification without omissions.

S.N	Emp ID	Full Name	Designation / Role	Department	Specific Platform Access & Responsibilities
1	**GOAG-A001**	**Yalamarthi Tulasi Ram**	Cofounder, Director	Executive Management	Simple Director dashboard: Financials, Sales, Production tracking, Marketing schedule, Vendor payment schedules.
2	**GOAG-A002**	**Bandaru Santhosh Kumar**	Cofounder, Director	Executive Management	Simple Director dashboard: Financials, Sales, Production tracking, Vendor management & purchase approvals.
3	**GOAG-A003**	**V Vishwateja Kumar Reddy**	Cofounder, Head Quality Control	QC & Procurement	Director dashboard: Financials, Sales, Production, Vendor management, Payment schedules, Custom quote approvals, Purchase lead.
4	**GOAG-A004**	**Battula Saran Kumar**	Cofounder, Head of Finance	Finance & Accounts	Director dashboard: Financials, Sales, Production, Store inventory tracking, Vendor management, Payments, Custom quote approvals, Tax admin.
5	**GOAG-A005**	**Ganesh Pentela**	Cofounder, Head - HR & Operations	Operations & HR	**Master access to entire ERP:** Director dashboard, Store inventory, Vendor management, Quote approvals, Leave approvals, full HR/Payroll admin.
6	**GOAG-A006**	**Vamsi Krishna K**	Cofounder, Head Aftersales	Executive & Aftersales	Director dashboard: Financials, Sales, Production, Store inventory tracking, Vendor management, Payment schedules, Quote approvals.
7	**GOAG-E007**	Kapil Kumar K	Production Manager	Production Operations	Sending quotes internally, Store inventory (view only), Task assignment, Component purchase requests, Leave/Payslip ESS access.
8	**GOAG-E008**	Mandugula Vamshi Teja	Manager - Sales	Commercial Sales	Customer data entry & tracking (lead to sale, delivery date confirmation), Marketing plan, Auto/Manual quotes, Leave/Payslip ESS.
9	**GOAG-E009**	Voleti Veera Babu	Lead Production, R&D	Production & R&D	Store inventory (view only), Task assignment to technicians, Component purchase requests, Leave/Payslip ESS, R&D workspace.
10	**GOAG-E010**	Meegala Vikas	Lead - Aftersales	Aftersales Support	Aftersales customer tracking (entry to delivery confirmation), Marketing plan, Auto/Manual quotes, Service diagnostics, Supervisor.
11	**GOAG-E011**	Guduri Harsha Deep	Supervisor - Sales & Purchase	Stores & Procurement	Store inventory tracking & daily inward/outward updates, Raising purchase requests, Leave/Payslip ESS.
12	**GOAG-E012**	Uppalapati Maneesha	Admin Assistant & Accountant	Accounts & HR Admin	Store inventory & inward/outward updation, Purchase requests, Accounts management (invoices & e-waybills), HR admin, Leave/Payslip ESS.
13	**GOAG-E013**	Darla Chandra Shekar	Design Engineer	Engineering & Design	Standard Employee Self-Service: Leave requests, assigned engineering tasks, request for payslips, regular employee dashboard.
14	**GOAG-E014**	Bommali Himagiri	Drone Technician	UAV Assembly & Fab	Standard ESS: Leave requests, assigned drone manufacturing tasks, 20-page checklists (e.g. GOAG-5X-001), payslip requests.
15	**GOAG-E015**	Bommali Yuvaraju	Senior Technician Drone Assembling	UAV Assembly	Standard ESS: Leave requests, assigned assembly tasks, executing checklists, payslip requests, regular employee dashboard.
16	**GOAG-E016**	Ghanta Sasidhar	Drone Technician	UAV Assembly	Standard ESS: Leave requests, assigned manufacturing tasks, executing product checklists, payslip requests, regular employee dashboard.
17	**GOAG-E017**	Kaluva Palle Venkatesh	Drone Technician	UAV Assembly	Standard ESS: Leave requests, assigned assembly tasks, executing product checklists, payslip requests, regular employee dashboard.
18	**GOAG-E018**	Jagana Adhi Lakshmi	Drone Technician	UAV Assembly	Standard ESS: Leave requests, assigned assembly tasks, executing product checklists, payslip requests, regular employee dashboard.
19	**GOAG-E019**	Vuppala Yashwanth Kumar	UAV Engineer	Avionics & Systems	Standard ESS: Leave requests, assigned technical tasks, executing assembly & flight checklists, payslip requests, regular dashboard.
20	**GOAG-E020**	Shaik Muzeeb	Drone Technician	UAV Assembly	Standard ESS: Leave requests, assigned assembly tasks, executing product checklists, payslip requests, regular employee dashboard.
21	**GOAG-E021**	Mandem Ashok Kumar	Sales Executive	Commercial Sales	Standard ESS: Leave requests, assigned sales tasks, request for payslip, regular employee dashboard access.
22	**GOAG-E022**	Sardena Vijay Kumar	Drone Technician	UAV Assembly	Standard ESS: Leave requests, assigned assembly tasks, executing product checklists, payslip requests, regular employee dashboard.
23	**GOAG-E023**	Thotakura Saida Rao	UAV Engineer	Avionics & Testing	Standard ESS: Leave requests, assigned engineering tasks, executing checklists, payslip requests, regular employee dashboard.

SECTION 3.2 • SECURITY & ACCESS GOVERNANCE

Departmental Role & Access Control Matrix

Formal operational authorization matrix across system modules. Indicators: View (V), Create (C), Edit (E), Approve (A), Manage (M), Master (*).

Role / Role Group	Sales & Leads	Quotes (Auto/Cust)	Work Orders	Checklists (20-Page)	Test Flight	QC & Images	Stores & Stock	Purchase Request	Purchase Orders	Invoices & E-Way	Petty Cash	HR & Payroll	Aftersales Support	Service Point (OTP)	Director Cockpit	Master Data
Directors (A001, A002, A006)	**V**	**A (Cust)**	**V**	**V**	**V**	**V**	**V**	**V**	**A**	**V**	**V**	**V**	**V**	**V**	**M**	**V**
Head of QC (A003 - Vishwa)	**V**	**A (Cust)**	**V**	**V**	**V**	**M**	**V**	**C / M**	**A / M**	**V**	**V**	**V**	**V**	**V**	**M**	**V**
Head of Finance (A004 - Saran)	**V**	**A (Cust)**	**V**	**V**	**V**	**V**	**V**	**V**	**A**	**M**	**M**	**V**	**V**	**V**	**M**	**E / M**
Master Admin (A005 - Ganesh)	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Production Mgr (E007 - Kapil)	-	C (Int)	M	M	V	V	V (Only)	C	-	-	C (ESS)	-	-	-	-	-
Sales Manager (E008 - Vamshi)	M	C / E	V (Stat)	-	-	-	-	-	-	V (PI)	C (ESS)	-	-	-	-	-
Lead Prod & R&D (E009 - Veera)	-	-	M	M	V	V	V (Only)	C	-	-	C (ESS)	-	-	-	-	-
Lead Aftersales (E010 - Vikas)	-	C / E	-	-	-	-	-	-	-	C (Svc)	C (ESS)	-	M	M (Diag)	-	-
Stores Supv (E011 - Harsha)	-	-	V (Pick)	-	-	-	M (In/Out)	C	-	-	C (ESS)	-	-	C (Inw)	-	-
Accountant (E012 - Maneesha)	-	-	-	-	-	-	M (In/Out)	C	V / Tag	M (Inv/EW)	M (Tag)	M (Admin)	-	-	-	E / M
Design Eng (E013 - Chandra)	-	-	-	-	-	-	-	-	-	-	C (ESS)	-	-	-	-	-
Drone Techs (E014-E018, E020, E022)	-	-	V (Assigned)	C / E	-	-	-	-	-	-	C (ESS)	-	-	-	-	-
UAV Engineers (E019, E023)	-	-	V (Assigned)	C / E	C / E	V	-	-	-	-	C (ESS)	-	-	E (Diag)	-	-
Sales Exec (E021 - Ashok)	C / E	C (Auto)	-	-	-	-	-	-	-	-	C (ESS)	-	-	-	-	-
Authorized Dealers (External)	-	*C (Req)*	*V (Own)*	-	-	-	-	-	-	*V (Own)*	-	-	*C (Tick)*	-	-	-
Drone Customers (External)	-	-	-	-	-	-	-	-	-	*V (Own)*	-	-	*C (Tick)*	*V (Own)*	-	-

CONFIRMED PERMISSION RULES:

- **Master Access:** Ganesh Pentela possesses full master administrative access across all platform modules, tables, and HR/Operations systems.
- **Master Data & Tax Edit:** Restricted exclusively to Uppalapati Maneesha, Battula Saran Kumar, and Ganesh Pentela.
- **Production Privacy:** Assembly technicians and UAV engineers are strictly prohibited from viewing sales values, financial terms, or margins on Work Orders.

DECISION REQUIRED — GRANULAR ACCESS POLICIES:

- Can Sales Executive Ashok Kumar issue custom discounted quotes, or are they restricted strictly to auto quotes based on list price?
- Can Senior Technician Drone Assembling Yuvaraju Bommali (GOAG-E015) sign off on checklists for other junior technicians?
- Can Production Managers view vendor part purchase costs, or only physical component availability?

SECTION 4 • FOUNDATION DATA LAYER

Master Data Architecture & GST Rule Engine

Centralized catalog management, Bill of Materials definitions, strict administrative editing access, and automated dynamic taxation.

Products & BOM Masters — CONFIRMED

Products Catalog: Complete drone models (e.g. GOAG 5X Drone, Graydon Drone), customized payloads, agricultural sprayer rigs, and ground control stations (GCS).
Engineering BOM: Multi-level Bill of Materials linking every product to its exact component pick-list, quantities, unit specs, and default suppliers.

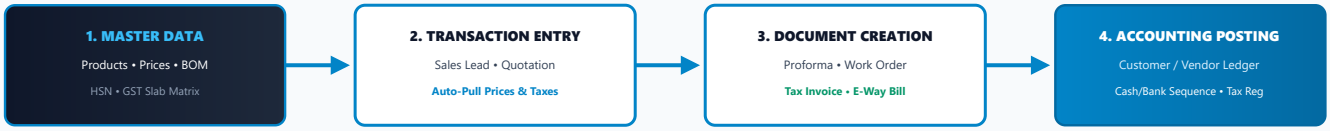
Components & Spares Master — CONFIRMED

Hardware Catalog: Motors, Electronic Speed Controllers (ESCs), carbon fiber propellers, flight controllers, GPS modules, smart batteries, battery chargers, telemetry radios, wiring harnesses.
Tracking: Master unit pricing, reorder thresholds, minimum lead times, and internal part numbers.

Tax & HSN Compliance Master — STRICT ACCESS

Taxation Rules: Harmonized System Nomenclature (HSN) codes mapped per item, standard GST slab percentages (e.g. 5%, 12%, 18%, 28%).
Strict Access Authorization: Master Data, catalog pricing, and tax parameters can *ONLY* be edited by: **Maneesha, Saran, and Ganesh**.

MASTER DATA PROPAGATION → TRANSACTION → DOCUMENT → ACCOUNTING LEDGER PIPELINE



BUSINESS REQUIREMENT: AUTOMATED GST RULE ENGINE

The system must eliminate manual tax calculation. Quotes and invoices must adapt dynamically to Indian GST statutory norms based on customer location and dispatch point:

- **Intrastate Sales (Same State):** System splits applicable GST equally into Central GST (CGST) and State GST (SGST).
- **Interstate Sales (Different State):** System applies Integrated GST (IGST) for the full rate.
- **Automated Line Extraction:** Each line item automatically pulls its registered HSN code and statutory tax percentage from the Master Catalog.

VERIFY BEFORE IMPLEMENTATION — TAX/GST NORMS:

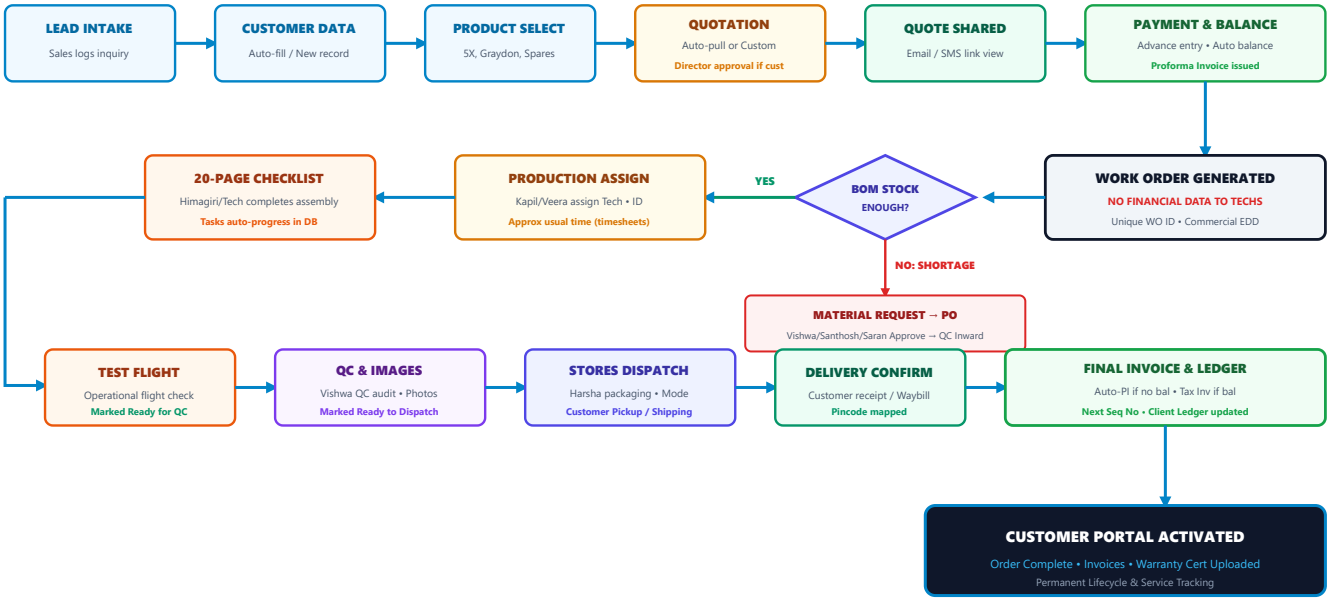
Technical & Legal Verification Action Item: While the business requirement dictates dynamic invoice adaptation, the exact statutory tax logic, place of supply rules, GSTIN validation APIs, and automated E-Way Bill JSON schemas must be formally audited and signed off by GOAG's practicing Chartered Accountant (CA) before backend database implementation begins.

SECTION 5 • CORE COMMERCIAL & FULFILLMENT PIPELINE

Complete Sales → Production → QC → Delivery Macro Flow

The enterprise lifecycle map tracking a drone order from initial inquiry to final customer portal activation across 9 distinct operational roles.

1. Customer Action | 2. Sales Action | 3. Management Approval | 4. Production Lead | 5. Assembly Technician | 6. QC Engineer | 7. Stores & Logistics | 8. Accounts Finance | 9. System Automation



CONFIRMED PIPELINE MANDATE:

Every unit manufactured follows this strict sequential chain. Work Orders cannot bypass Stores BOM validation; assembly cannot bypass the 20-page checklist; dispatches cannot bypass QC image uploads; and Customer Portals receive permanent documentation upon delivery.

SECTION 6 • COMMERCIAL FRONT-END

Sales Intake, Quotations & Delivery Commitments

Lead capture, customer data enrichment, dual-mode quotation generation, digital quote sharing, and estimated delivery scheduling.

1. Lead Intake & Profiling — CONFIRMED

Responsible: Mandugula Vamshi Teja (Manager - Sales) and Mandem Ashok Kumar (Sales Executive).
Actions: Intake inquiry from web, dealer, phone, or expo. System validates phone number; if existing customer, links historical record; if new, creates customer record (Name, Contact, Company, Delivery Pincode, GSTIN).

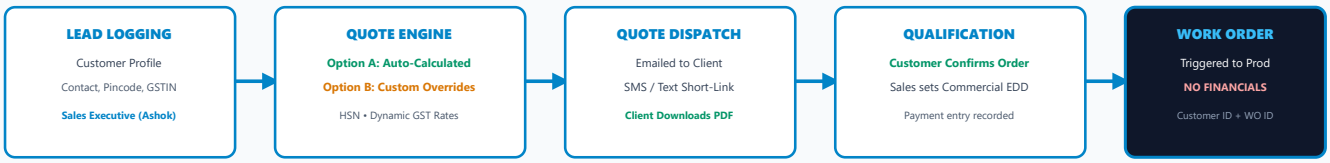
2. Dual-Mode Quotations — CONFIRMED

Mode A: Prefilled / Automatic: System auto-pulls catalog unit prices, HSN codes, and GST rates from Master Data. No manual math allowed.
Mode B: Custom Quotation: Allows manual price adjustments or special discounts. Must route for Director approval (Tulasi Ram / Saran / Vishwateja).

3. Multi-Channel Sharing — CONFIRMED

Client Delivery: Formatted professional quote generated with unique Quote ID. Automatically emailed to client or dispatched via SMS/Text containing a secure, short link allowing instant online viewing and PDF download.

SALES LEAD → QUOTATION → QUALIFICATION → WORK ORDER TRIGGER FLOW



⚖️ CRITICAL OPERATIONAL DISTINCTION: COMMERCIAL EDD VS. EMPLOYEE WORK-TIME TRACKING

1. Commercial Estimated Delivery Date (EDD): Set by Sales (Vamshi Teja) upon quote qualification. This represents the customer-facing delivery commitment based on production queue and parts availability.
2. Internal Employee Work-Time Tracking: When Production Managers assign an assembly task to a technician (e.g. Himagiri), they specify the "*approximate usual time required by that employee*". **This is NOT a production deadline, SLA, or punitive target.** Its explicit purpose is to record timesheets, understand employee work patterns, monitor productivity, and build internal discipline. It must never be shown to customers or treated as a guaranteed completion deadline.

SECTION 7 • COMMERCIAL-TO-OPERATIONS GATEWAY

Quotation → Payment → Work Order Transition

Financial recording, automated balance calculation, proforma issuance, Work Order creation, and strict financial segregation.

1. Quote Accepted
Client confirms quotation via email, text link, or signed proposal. Sales updates status to *Confirmed Order*.

2. Payment Entry
Sales/Accounts logs advance payment received (NEFT, UPI, Cheque, Cash). System auto-calculates remaining balance.

3. Proforma (PI)
Proforma Invoice is generated reflecting advance received, remaining balance, and terms. Dispatched to client.

4. Work Order (WO)
System automatically spawns Work Order with unique WO ID and linked Customer ID. Routed to Production Manager.

WORK ORDER DATA STRUCTURE & PRIVACY BOUNDARY

(Generated Work Order Artifact) — STRICT ACCESS GOVERNANCE

✓ VISIBLE TO PRODUCTION TEAMS (TECHNICIANS & ENGINEERS)

Work Order ID:	**WO-2026-05X-014**
Customer Reference:	CUST-8492 (Customer Name & Contact)
Product Ordered:	GOAG 5X Heavy-Lift Drone (Qty: 1)
BOM Component Spec:	Standard 5X Quad Rigs + Dual Smart Battery
Commercial EDD:	18-Oct-2026 (For target staging)
Assigned Checklist:	CL-5X-PROD (20-Page Database Form)

⃠ STRICTLY RESTRICTED / REDACTED FROM PRODUCTION

Item Selling Price:	[REDACTED • CONFIDENTIAL]
Advance Payment Received:	[REDACTED • CONFIDENTIAL]
Outstanding Balance:	[REDACTED • CONFIDENTIAL]
Discounts / Margins:	[REDACTED • CONFIDENTIAL]
Commercial Terms / Ledger:	[REDACTED • CONFIDENTIAL]

STAKEHOLDER WORK ORDER VISIBILITY MATRIX

Stakeholder Group	Key Named Individuals	Work Order Visibility Level & Access Scope
Executive Directors	Tulasi Ram, Santhosh, Vishwateja, Saran, Ganesh, Vamsi	**Full Unrestricted Visibility:** Full financial terms, customer billing, BOM specifications, assembly status, and profit margins.
Production Management	Kapil Kumar (Mgr), Voleti Veera Babu (Lead)	**Technical & Operational Management:** Full access to technical specifications, component pick-lists, technician assignment; zero financial figures.
Sales Management	Mandugula Vamshi Teja (Manager - Sales)	**Status Visibility Only:** Can track manufacturing milestone status (e.g. WIP, Test Flight, QC) to update the client; cannot alter technical specs.
Assembly Technicians	Himagiri, Yuvaraju, Sasidhar, Venkatesh, etc.	**Assigned Unit Checklist Only:** Access limited strictly to their designated drone unit checklist and build notes; zero commercial visibility.

CONFIRMED DELAYED PAYMENT GOVERNANCE:
Under normal operations, a final tax invoice is only issued upon 100% payment clearance. If a commercial agreement permits delayed/credit payment, explicit approval from the **Core Management Team (Directors)** must be recorded in the system before production or dispatch is released.

SECTION 8 • CUSTOMER LIFECYCLE & EXTERNAL ENGAGEMENT

Customer Account Provisioning & Portal Architecture

Automated account provisioning, customer entity binding, dual-path registration, and unified digital portal services.

Path A: Brand New Customer — CONFIRMED

Trigger: First-time purchase order confirmed by Sales.
Automated System Actions:

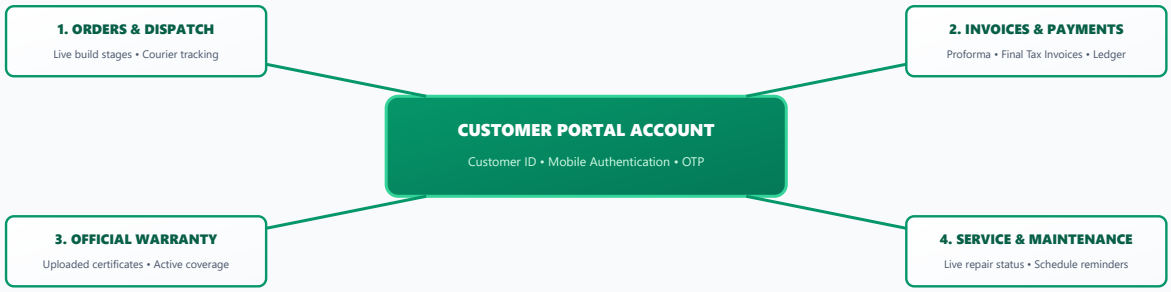
- Generates a globally unique `Customer_ID`.
- Provisions secure Customer Portal account credentials.
- Dispatches automated welcome email and SMS onboarding link.
- Binds initial Work Order, Proforma Invoice, and delivery profile.

Path B: Existing Registered Customer — CONFIRMED

Trigger: Repeat drone purchase or spare parts order.
Automated System Actions:

- Matches primary phone number / email against Master Customer Registry.
- Appends new Work Order, quotation, and payment records to existing account.
- No duplicate accounts created; maintains consolidated purchase ledger.
- Instantly surfaces live progress inside client's active portal dashboard.

CUSTOMER PORTAL UNIFIED DATA & SERVICES ARCHITECTURE

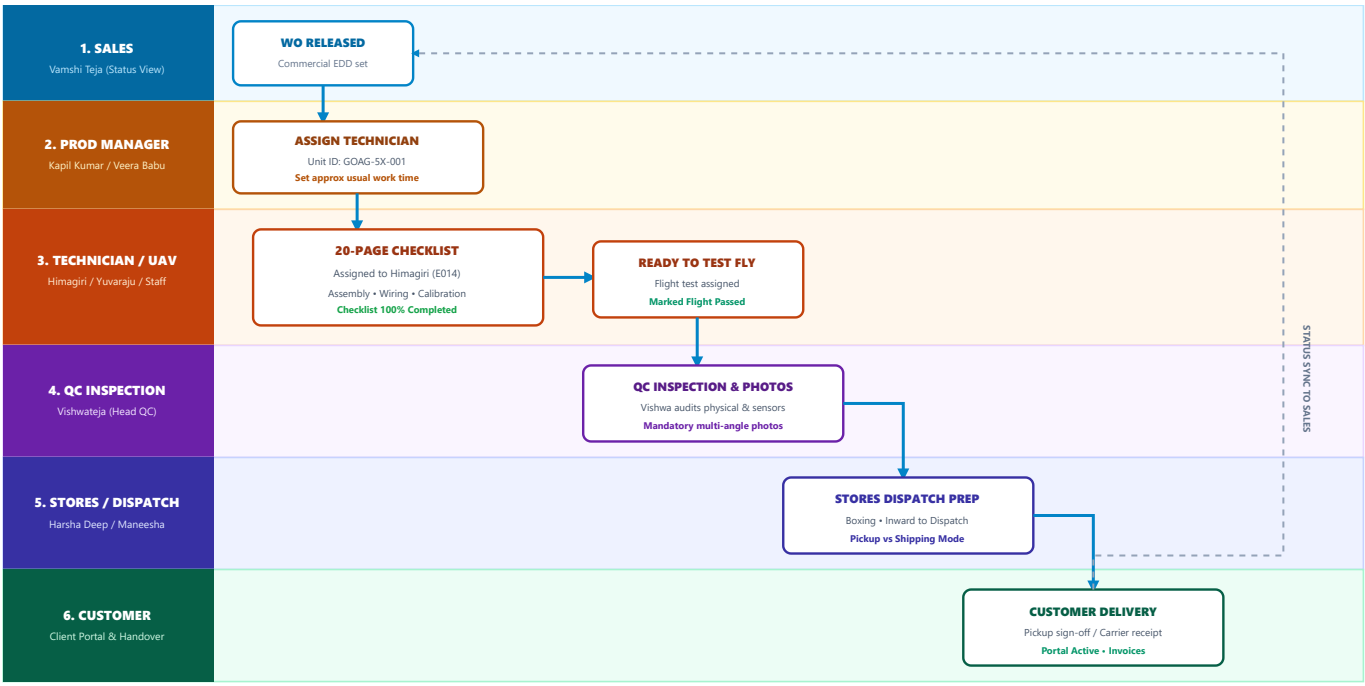


CONFIRMED PORTAL INTEGRITY REQUIREMENT:
The Customer Portal is not merely a tracking screen; it is the permanent legal repository for the client's drone fleet. GOAG administrators upload official signed warranty certificates and tax invoices directly into this portal, ensuring customers maintain continuous access to compliance documentation throughout the operating life of their aircraft.

SECTION 9 • MANUFACTURING & ASSEMBLY OPERATIONS

Drone Production & Assembly 6-Lane Swimlane

Cross-departmental handoffs from Work Order receipt to final Stores dispatch, featuring the 20-page checklist and unit tracking.



CONFIRMED WORKFLOW EXAMPLE • GOAG-5X-001:
When Himagiri is assigned Work Order `GOAG-5X-001` (the 5X heavy-lift drone), the 20-page checklist is auto-bound to his dashboard with his approximate usual build duration recorded for timesheet analysis. Completion dynamically unlocks the test flight queue.

SECTION 10 • OPERATIONAL QUALITY ENFORCEMENT

Database-Driven 20-Page Checklist Architecture

Digitalizing complex aerospace assembly guidelines into interactive, uniquely identifiable, database-backed inspection stages.

Database-Driven Architecture CONFIRMED

Checklists are not static paper or PDFs. They are dynamic database models. Each production unit instantiates a unique `Checklist_ID` linked permanently to `WO_ID`, aircraft serial, and the assigned technician ID.

The 20-Page Drone Checklist CONFIRMED

Explicitly mandated by source specification for complex drones (such as the GOAG 5X). Contains 20 comprehensive sections covering mechanical, electronic, propulsion, firmware, and pre-flight calibrations.

Automated Task Progression CONFIRMED

When a technician verifies and signs off on a checklist stage, the platform automatically logs timestamps, stores parameter readings, and dynamically loads the subsequent task into the technician's pending work queue.

STRUCTURE OF THE 20-PAGE MANUFACTURING & ASSEMBLY CHECKLIST

Stage	Assembly Domain	Critical Verification & Parametric Test Checks	Technician Action
Pages 1–3	Airframe & Structural Hardware	Carbon tube alignment, motor mount torque check, fold-joint rigidity, landing gear dampening.	Torque verification & visual sign-off
Pages 4–6	Propulsion & Power Harness	ESC wiring polarity, solder joint resistance, motor spin orientation, heat sink thermal paste.	Solder inspection & continuity check
Pages 7–9	Avionics & Flight Controller	Flight controller vibration isolation, PDB power distribution check, BEC 5V/12V rail testing.	Multimeter voltage log
Pages 10–12	Sensors, Telemetry & GPS	Dual GPS compass offset, RF telemetry link strength, lidar rangefinder calibration.	GCS telemetry calibration log
Pages 13–15	Payload & Application Systems	Agricultural pump flow rate, spray nozzle pattern, tank level sensor, payload gimbal stabilization.	Functional wet/pressure test
Pages 16–17	Firmware & Ground Station	Flashing stable flight firmware, PID parameter loading, RC failsafe & RTH parameter locking.	Firmware version & hash recording
Pages 18–19	Ground Bench & Safety Audit	Arming switch safety check, prop-off motor test, battery bay lock engagement, emergency cutoff.	Arming & failsafe verification
Page 20	**Pre-Flight Readiness Sign-Off**	**Comprehensive technician build sign-off, tool inventory check, marking unit "Ready for Test Flight".**	**Digital signature & stage commit**

STAGE COMPLETION & WORKFLOW HANDOFF MECHANISM

Technician completes Page 20 → Status changes to **"Ready to Test Fly"** → Unit automatically surfaces on Production Manager's cockpit → Manager assigns flight pilot → Flight task pops up on pilot's dashboard with link to Flight Checklist.

CONFIRMED ARCHITECTURAL VALUE:

This database architecture guarantees complete build traceability. If a drone encounters an issue in the field months later, engineers can inspect the exact technician remarks, torque calibrations, and component serials logged during assembly.

SECTIONS 11 & 12 • VERIFICATION & ASSURANCE GATEWAYS

Test Flight Protocol & Quality Control (QC) Architecture

Operational flight validation, certified pilot sign-off, comprehensive physical audit, and mandatory photographic evidence capture.

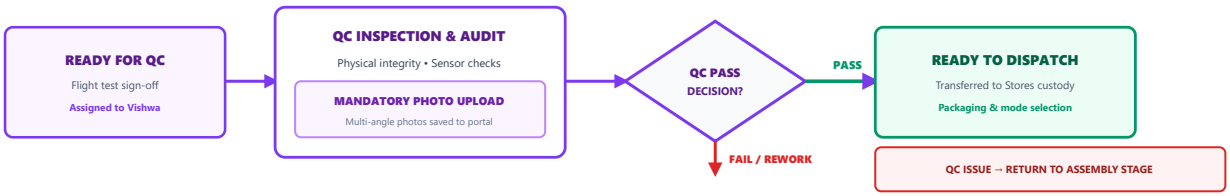
SECTION 11: TEST FLIGHT PROTOCOL & PROGRESSION

CONFIRMED REQUIREMENT

1	2	3	4
Assembly Done	**Flight Queued**	**Flight Testing**	**Flight Passed**
20-page checklist 100% completed by technician.	Manager marks unit "Ready for Test Flight".	Pilot verifies hover, navigation, failsafe RTH.	Unit automatically transitions to "Ready for QC".

SECTION 12: QUALITY CONTROL (QC) PROCESS & EVIDENCE CAPTURE

CONFIRMED REQUIREMENT



DECISION REQUIRED — EXACT QC REJECTION & REWORK ROUTING PROTOCOL:

The source documents confirm that QC failure halts dispatch and marks a QC issue. However, GOAG must formally decide:

- Does a rejected unit automatically return to the original assembling technician (e.g. Himagiri), or does the Production Manager reassign it?
- Does a failed drone require a complete re-run of all 20 checklist pages, or only the specific failed subsystem?
- Is a second test flight mandatory after mechanical/electronic rework, or can QC clear minor cosmetic fixes on the bench?

SECTION 13 • SUPPLY CHAIN & MATERIALS MANAGEMENT

Stores Inventory & Automated BOM Availability Engine

Real-time Bill-of-Materials stock validation, automated shortage detection, daily store inward/outward logs, and mandatory pre-stock QC.

Automated BOM Checking — CONFIRMED

When a Work Order is spawned (e.g. 1× 5X Drone + 1× Graydon), the engine parses each product's engineering BOM and performs a millisecond query against live store inventory to verify component availability.

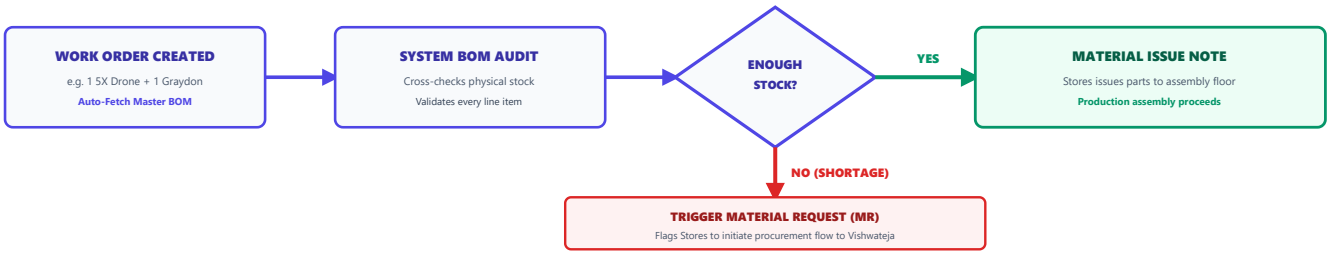
Instant Shortage Flagging — CONFIRMED

If even a single screw, ESC, or telemetry radio is deficient, the system immediately flags Stores (Harsha Deep / Maneesha) to generate an automated **Material Request (MR)** routed to procurement.

Mandatory Pre-Stock QC — CONFIRMED

Storekeepers cannot place received goods directly into available stock. All raw materials and vendor deliveries must pass physical QC inspection before being posted to active inventory.

AUTOMATED BOM INVENTORY VALIDATION & MATERIAL ISSUANCE FLOW



DAILY STORES INWARD & OUTWARD RESPONSIBILITIES (HARSHA DEEP / MANEESHA)

Store Operation	Action & Physical Flow	Mandatory System Control & Quality Gateway
Daily Component Inward	Receiving shipments from external component vendors.	**Mandatory Inward QC:** Components cannot be stocked until QC inspects and approves incoming items.
Daily Component Outward	Issuing kits, airframes, motors, and screws to production floor.	**Work Order Requisition:** Parts only released against verified WO and Material Issue Note.
Finished Product Outward	Receiving certified drones from QC into dispatch staging.	**Mandatory Outward Verification:** Packing slip generated; verification of customer pickup vs courier mode.

SECTION 14 • PROCUREMENT & SUPPLIER MANAGEMENT

Purchase Request & Vendor Management Architecture

Automated shortage conversion, dedicated procurement leadership, executive 3-person approval gate, and alternate vendor remarks.

Purchase Leadership — CONFIRMED

Procurement Lead: V Vishwateja Kumar Reddy (GOAG-A003) manages primary vendor negotiations and supplier relations. Material requests from Stores route directly to his dashboard.

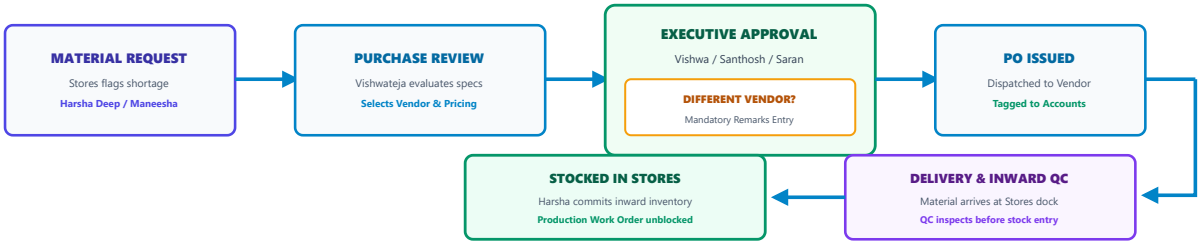
3-Person Approval Authority — CONFIRMED

Authorizers: Vishwateja, Bandaru Santhosh Kumar (GOAG-A002), or Battula Saran Kumar (GOAG-A004). Any one of these three directors possesses authorized signing authority to approve purchase orders.

Alternate Vendor Rule — CONFIRMED

Mandatory Justification: If components are procured from an alternate or non-primary vendor (due to stockouts or pricing), the approver *must* enter mandatory remarks into the system before PO release.

END-TO-END COMPONENT PROCUREMENT & INWARD QC LIFECYCLE



CONFIRMED PROCUREMENT GOVERNANCE:

Purchase requests cannot be converted to Purchase Orders by general staff. Only Vishwateja, Santhosh, or Saran can execute approvals. The system permanently logs which director signed off and preserves alternate vendor justification remarks for audit.

SECTION 15 • FINANCIAL OPERATIONS & LEDGER AUTOMATION

Purchase Tagging & Director Financial Visibility

Real-time routing of procurement liabilities into accounts, automated expense ledger posting, and multi-temporal cashflow governance.

Automatic Accounts Tagging — CONFIRMED

The moment a PO is approved by Vishwa/Santhosh/Saran, the system automatically routes and tags it onto the financial worklists of: **Uppalapati Maneesha (GOAG-E012)**, **Battula Saran Kumar (GOAG-A004)**, and future accounts personnel.

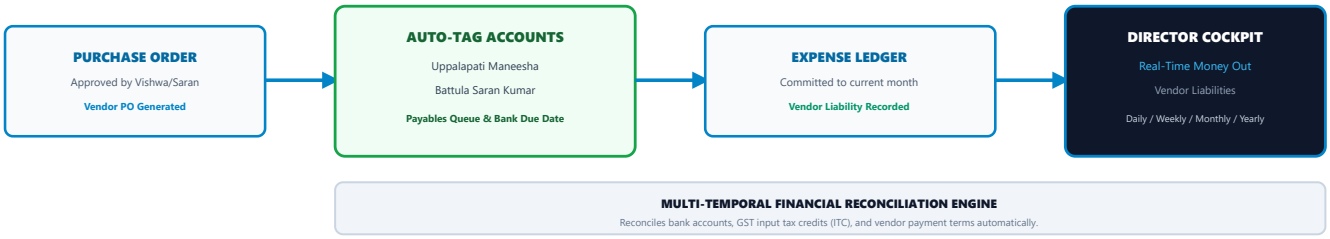
Monthly Expense Auto-Logging — CONFIRMED

Approved PO values are instantly committed as payables in the current month's company expense ledger. Accounts staff do not need to manually re-type purchase order invoices into financial books.

Real-Time Cashflow Visibility — CONFIRMED

Executive Directors (Tulasi Ram, Santhosh, Vishwateja, Saran, Ganesh, Vamsi) receive live financial visibility into total **Money In** and **Money Out** across **Day, Week, Month, and Year** views.

PURCHASE ORDER TAGGING → EXPENSE LEDGER → EXECUTIVE COCKPIT FLOW



CONFIRMED FINANCIAL TRANSPARENCY REQUIREMENT:

There can be no “off-the-books” purchase orders. The moment procurement commits to a supplier, the liability immediately registers inside the executive dashboard. Directors know the exact cash obligations of the company for any given day or month in real time.

SECTION 16 • OPERATIONAL EXPENSE MANAGEMENT

Dual-Path Petty Cash & Voucher Management

Accounts-initiated disbursements vs employee self-service reimbursement claims with mandatory digital receipt tagging.

PATH A: ACCOUNTS-INITIATED PETTY CASH

CONFIRMED

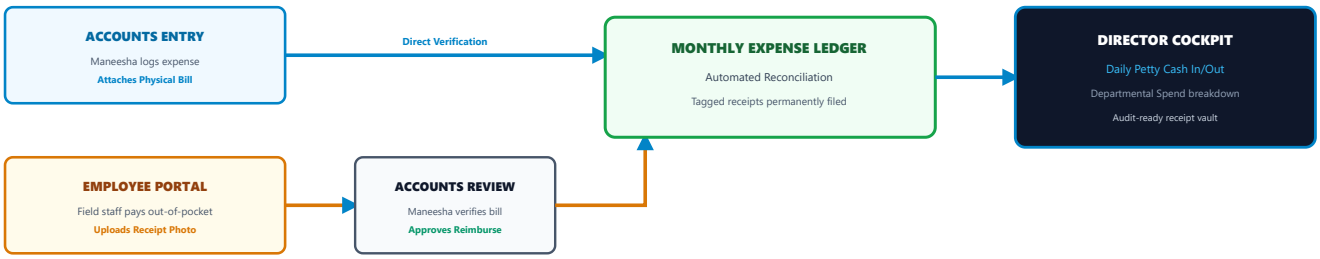
Originator: Accountant Uppalapati Maneesha (GOAG-E012).
Workflow: Accountant pays for operational consumables (stationery, workshop cleaning supplies, local freight), enters the amount directly into the petty cash ledger, tags the scanned receipt/bill, and commits the entry to the monthly expense ledger.

PATH B: EMPLOYEE-INITIATED EXPENSE CLAIM

CONFIRMED

Originator: Any staff member (Technician, UAV Engineer, Sales, Pilot).
Workflow: Employee pays an out-of-pocket operational cost (hardware fasteners, field travel, test fuel), logs the expense in their Employee Portal, uploads a photo of the bill. Accounts receives the claim, verifies the receipt, approves reimbursement, and posts it to the monthly expense ledger.

DUAL-PATH PETTY CASH LOGGING & MONTHLY EXPENSE RECONCILIATION



CONFIRMED PETTY CASH VOUCHER RULE:

No petty cash claim can be committed to the ledger without an attached digital image of the receipt or voucher. This dual-path architecture ensures field staff are reimbursed rapidly while preventing untracked cash leakages across company operations.

SECTION 17 • FINANCIAL DOCUMENTATION & STATUTORY COMPLIANCE

Invoicing Architecture & Cash vs Account Separation

Multi-category billing, strict sequential isolation between Cash and Bank transactions, and administrative finalization controls.

1. Sales Invoices — CONFIRMED

Scope: Complete aircraft systems (GOAG 5X, Graydon), customized agricultural rigs, ground telemetry kits, and multi-unit fleet purchases. Generated upon delivery fulfillment.

2. Spares Invoices — CONFIRMED

Scope: Aftermarket component sales (motors, carbon props, ESCs, smart batteries, chargers, nozzles) purchased by dealers or walk-in customers.

3. Custom Invoices — CONFIRMED

Scope: Specialized engineering consulting, non-standard drone modifications, pilot training programs, and bespoke enterprise contracts.

MANDATORY SEPARATION: CASH SEQUENCE VS. ACCOUNT (BANK) SEQUENCE

CONFIRMED SOURCE REQUIREMENT

A. CASH TRANSACTION SEQUENCE

Transaction Type: Physical currency payments received at branch or counter.
Numbering Engine: Dedicated, independent sequence (e.g. `GOAG-CASH-INV-YYYY-XXXX`).
Ledger Destination: Directly credits the Physical Cash Book; strictly segregated from banking registers.
Statutory Rule: Complies with cash receipt limit thresholds.

B. ACCOUNT (BANK) TRANSACTION SEQUENCE

Transaction Type: Electronic bank transfers (NEFT, RTGS, IMPS, UPI, Corporate Cheques).
Numbering Engine: Dedicated, independent sequence (e.g. `GOAG-ACCT-INV-YYYY-XXXX`).
Ledger Destination: Credits designated Current Account ledgers; reconciles directly with bank statements.

STRICT FINAL INVOICE ISSUANCE GOVERNANCE

A final invoice cannot be issued speculatively. System strictly blocks invoice generation until all mandatory billing data (Customer legal entity, billing address, shipping address, valid GSTIN/PAN, and full component line items) is fully committed.
Authorized Admin Access: Restricted strictly to **Maneesha, Saran, and Ganesh**.

STATUTORY E-WAY BILL GENERATION

For dispatches exceeding statutory value limits (e.g. >₹50,000 consignment value under Indian GST), Accountant Maneesha generates compliant E-Way Bills incorporating vehicle/transporter details. Generated E-Way Bill numbers are bound to the final invoice and dispatch manifest.

SECTION 18 • COMMERCIAL SETTLEMENT & BILLING RULES

Payment, Proforma & Delivery Trigger Logic

Commercial settlement branches, credit approvals for delayed payments, and formal resolution of source-text billing conflicts.

Case 1: Full Payment Cleared — CONFIRMED

Condition: Outstanding balance is zero (100% advance or balance settled prior to dispatch).
Fulfillment: Dispatch hold released; drone cleared for customer pickup or courier shipping; document package dispatched; client ledger settled.

Case 2: Balance Outstanding — CONFIRMED

Condition: Partial advance received; client owes remaining balance upon manufacturing completion.
Fulfillment: Drone held in Stores staging pending final financial clearance, unless commercial credit approval is obtained.

Case 3: Approved Delayed Payment — CONFIRMED

Condition: Institutional or enterprise client requests credit terms (e.g. 15-day / 30-day payment cycle).
Mandatory Requirement: Requires explicit digital approval from the **Core Management Team (Directors)**.

🔴 CRITICAL DECISION REQUIRED — FINAL INVOICE / PROFORMA (PI) DISPATCH TRIGGER

A direct textual contradiction exists in the source operational notes regarding the document triggered upon delivery dispatch. **Per the Source-of-Truth rule, this must be decided by GOAG leadership before software development begins:**

Statement A (Source Section 3, Lines 14–17):
"The final invoice will only be shared if the full payment is cleared or getting approval from the core management team from their accounts (if they have agreed for some delayed payments which the core team has approved)..."
→ Represents standard commercial practice: Tax Invoice issued when fully paid or credit approved.

Statement B (Source Section 3, Lines 35–38):
"...and once marked delivered as it reached the client or dispatched then the auto PI must be emailed to the client if no balance and if balance is there then invoice with appropriate invoice number (next invoice number) and entered automatically into our ledger..."
→ Reverses terminology: Auto Proforma (PI) if no balance, and Tax Invoice if balance remains.

PROPOSED RESOLUTION FOR GOAG BOARD APPROVAL: Issue Proforma Invoice (PI) at quote acceptance → Issue Final GST Tax Invoice upon physical dispatch/delivery, regardless of balance, while carrying forward any unpaid balance as an Accounts Receivable debit in the Client Ledger.

COMMERCIAL RELEASE & MANAGEMENT APPROVAL GATE

Step 1: Manufacturing Finished → **Step 2:** Check Ledger Balance → **Step 3:** If Unpaid → Route to Core Management for Delayed Payment Approval → **Step 4:** Release Dispatch to Stores

SECTION 19 • FULFILLMENT & LOGISTICS EXECUTION

Dispatch, Logistics & Customer Delivery Architecture

Stores dispatch staging, multi-modal fulfillment selection (Customer Pickup vs Courier Shipping), ledger updates, and portal activation.

Stores Dispatch Custody — CONFIRMED

Responsible: Guduri Harsha Deep (GOAG-E011) and Uppalapati Maneesha (GOAG-E012). Stores takes physical and system custody of completed drones marked "Ready to Dispatch" by QC.

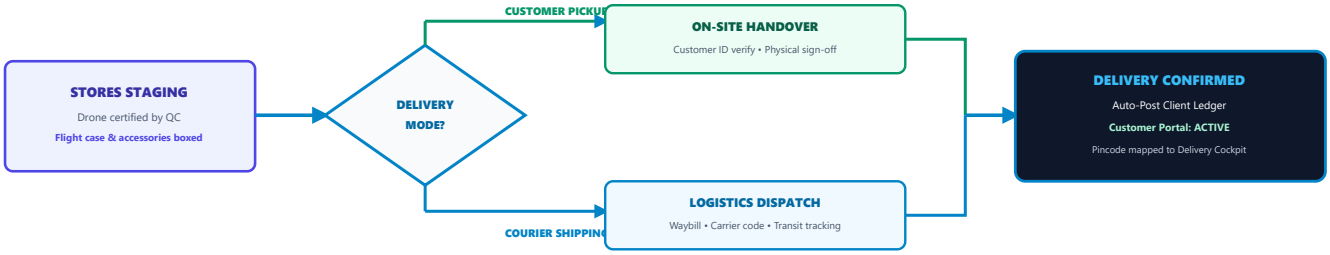
Dual Delivery Modes — CONFIRMED

Mode A: Customer Pickup: Direct physical collection at GOAG facility with customer verification.
Mode B: Logistics Shipping: Packaging in flight cases, courier dispatch, tracking code entry, and transit monitoring.

Automated Downstream Sync — CONFIRMED

Delivery sign-off triggers automated posting to the Client Ledger, assigns the next sequential tax invoice number, provisions permanent warranty docs, and logs destination pincode for the Director Delivery Map.

STORES FULFILLMENT MODE DECISION & LEDGER INTEGRATION LIFECYCLE



CONFIRMED GEOGRAPHIC PINCODE TRACKING:
During the dispatch recording stage, the system extracts the customer's delivery destination postal PIN Code. This data directly feeds the executive **Delivery Map** on the Director Dashboard, plotting visual pins of GOAG's operational aircraft footprint across the country.

SECTION 20 • AFTERSALES SUPPORT OPERATIONS

Inbound Support, Phone Resolution & Defect Analytics

Instant caller identification, first-contact resolution workflows, service point triage, and systematic defect pattern analysis.

Instant Caller Lookup — CONFIRMED

When an operator calls support, the technician enters their phone number. The platform instantly surfaces the customer's fleet profile, purchased drone models, purchase dates, warranty validity, and past service logs.

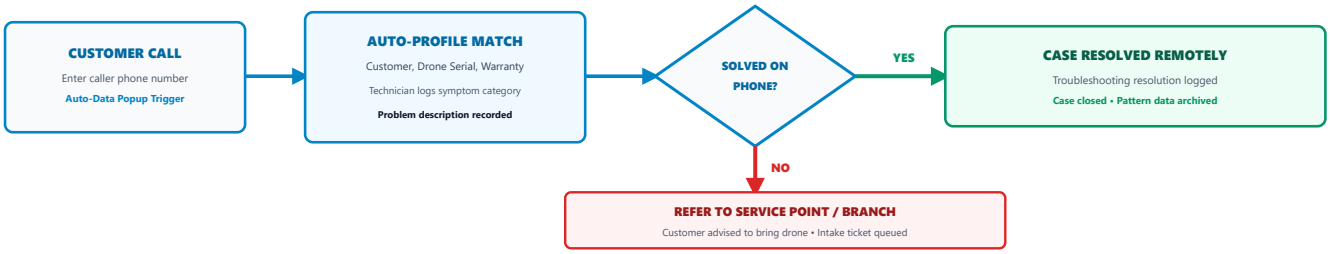
First-Contact Phone Triage — CONFIRMED

Technician logs the exact symptom and troubleshooting steps. If resolved over the phone (e.g. compass recalibration, firmware toggle), the resolution is logged and the ticket closed immediately.

Defect Pattern Analytics — CONFIRMED

All support logs are aggregated in a central analytics engine to detect recurring patterns (e.g. repeated ESC heating, battery BMS errors), providing direct statistical feedback to R&D and production.

INBOUND SUPPORT INTAKE & REMOTE VS PHYSICAL RESOLUTION ARCHITECTURE



CONFIRMED REASONING FOR SYSTEMATIC DATA LOGGING:

Even if a call lasts only 60 seconds, recording the issue type is mandatory. The architecture aggregates these records to detect fleet-wide failure trends. For example, if 15 customers report motor stutter after 20 flight hours, the platform flags a quality alert to QC Head Vishwateja and R&D Lead Veera Babu for component re-evaluation.

SECTIONS 21 & 22.1 • SERVICE DESK & PHYSICAL INTAKE

Support Staff Rotation & Service Point Intake Protocol

Rotational desk scheduling, physical stores intake, photographic damage documentation, and customer OTP authentication.

SECTION 21: AFTERSALES SUPPORT STAFF ROTATION

CONFIRMED REQUIREMENT

Confirmed Operational Concept: Support desk technicians are scheduled on a rotational basis, assigned on the previous day for the next day's shift. This ensures continuous coverage without overburdening individual technicians.

DECISION REQUIRED — ROTATION ALGORITHM:
GOAG must decide: Are all 9 drone assembly technicians part of the support rotation pool, or is rotation restricted to designated customer-facing personnel (Ashok, Vikas, Harsha)? Is scheduling automated or managed by Vikas?

SECTION 22.1: BRANCH SERVICE POINT PHYSICAL INTAKE & OTP VERIFICATION

CONFIRMED REQUIREMENT

1. **Stores Inward** — Stores receives physical drone at branch counter.
2. **Condition Photos** — Staff captures multi-angle damage and wear photos.
3. **Catalog Spares** — Logs all accessories brought (props, batteries, GCS).
4. **Customer OTP** — OTP verify • Links case to Customer Portal.

SERVICE INWARD DIGITAL ARTIFACT SPECIFICATION

Artifact Element	Recorded Field Parameters	Operational & Legal Purpose
Unique Service ID	e.g. `SRV-2026-BR01-0842`	Primary tracking entity throughout diagnostics, parts replacement, and billing.
Intake Photographic Evidence	High-res camera upload (airframe, motors, scratches, landing gear)	Protects GOAG against false claims of facility damage while in custody.
Accessories Brought Checklist	Batteries (Serials), Remote Controller, Charger, Props, Spray Nozzles	Prevents missing accessory disputes during final customer collection.
Customer Mobile OTP	**4-Digit / 6-Digit dynamic SMS OTP entered into system**	**Confirms customer consent and securely binds service tracking to client portal.**

CONFIRMED CUSTOMER PORTAL SERVICE INTEGRATION:
The moment the Customer OTP is verified at the branch counter, the service ticket immediately transitions onto the floor and displays live status on the customer's portal. Customers can monitor when diagnostics start, view technician remarks, and track estimated completion times in real time.

SECTION 22.2 • WORKSHOP SERVICE EXECUTION

Diagnostics, Service Checklists & Handover OTP

Supervisory allocation by Vikas, technician diagnostics, operational timesheet tracking, service remarks, and secure OTP release.

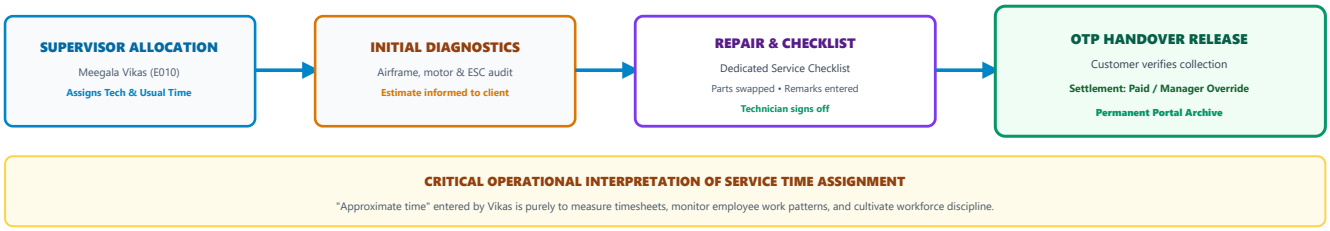
1. Vikas Assignment
Supervisor Meegala Vikas reviews intake photos, assigns ticket to designated UAV technician, and queues diagnostics.

2. Diagnostics & Cost
Technician logs faulty parts; Vikas communicates estimated repair cost and approximate usual service duration to customer.

3. Service Checklist
Dedicated service checklist auto-loaded. Technician executes repair, records part serials, and logs detailed work remarks.

4. OTP Handover
Upon payment clearance or manager override, final customer handover requires OTP verification. Ticket closed and archived.

IN-BRANCH TECHNICAL REPAIR & HANDOVER WORKFLOW



CRITICAL OPERATIONAL INTERPRETATION OF SERVICE TIME ASSIGNMENT

"Approximate time" entered by Vikas is purely to measure timesheets, monitor employee work patterns, and cultivate workforce discipline.

CONFIRMED DUAL-OTP SECURITY PROTOCOL:
Every physical branch service requires two independent OTPs: **(1) Intake OTP** to accept the drone into company custody, catalog pre-existing scratches, and link live status to the user portal; and **(2) Handover OTP** to release the repaired aircraft back to the owner once payment or manager warranty override is validated.

SECTIONS 23 & 24 • SERVICE SETTLEMENT & OUTSOURCED REPAIRS

Manager Warranty Overrides & External Service Centres

Commercial repair fee waivers, warranty validations, third-party vendor tracking for batteries/chargers, and logistics loops.

SECTION 23: SERVICE SETTLEMENT & MANAGER OVERRIDE GATE

CONFIRMED REQUIREMENT

Path A: Billable Technical Repair

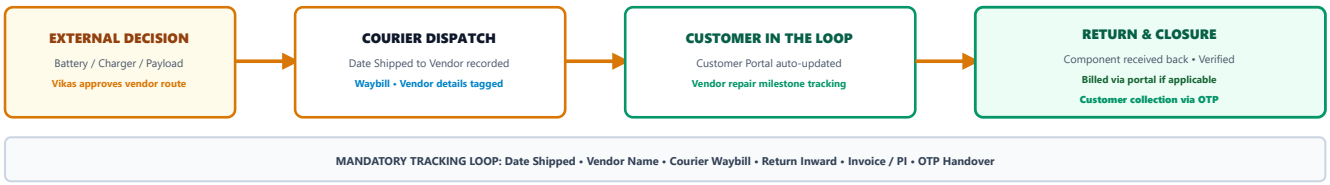
Customer accepts quotation → Payment logged in system by Aftersales / Accounts → Spares/Service Invoice generated → Handover verified via Customer OTP → Ticket archived.

Path B: Warranty Claim or Commercial Discount

Repair fee waived under warranty or commercial goodwill → **Aftersales Manager Override** applied by Meegala Vikas with mandatory justification remarks → Handover verified via Customer OTP.

SECTION 24: EXTERNAL SERVICE CENTRE LIFECYCLE (BATTERIES & CHARGERS)

CONFIRMED REQUIREMENT



CONFIRMED EXTERNAL SERVICE INTEGRITY:

Specialized components like high-capacity lithium flight batteries and smart multi-channel chargers cannot be repaired internally without OEM calibration rigs. The system maintains an unbroken chain of custody, documenting the exact date shipped to the vendor and synchronizing status with the customer's portal so the owner is never left guessing.

SECTION 25 • HUMAN CAPITAL & ATTENDANCE OPERATIONS

Human Resources Architecture & Attendance Tracking

Centralized administrative control by Ganesh and Maneesha, manual attendance with key-card roadmap, and salary sheet generation.

Exclusive HR Administrative Authority

STRICT ACCESS

Administrators: Ganesh Pentela (GOAG-A005) and **Uppalapati Maneesha (GOAG-E012)**.
Governance Mandate: Only Ganesh and Maneesha possess admin-level access to the complete HR bundle (employee profiles, salary master data, attendance regularization, leave approvals, payroll generation, and statutory records).

Employee Profile & Compensation Master

CONFIRMED

Master Record: Permanent repository for each team member's full profile, KYC, statutory IDs, banking details, formal Job Description, base compensation structure, and reporting line.

ATTENDANCE LOGGING ARCHITECTURE: PHASED IMPLEMENTATION

SOURCE CONFIRMED TIMELINE

PHASE 1: CURRENT IMPLEMENTATION (MANUAL LOGGING)

HR administration (Maneesha / Ganesh) performs daily manual entry into the platform. Employees are classified under 5 official attendance status codes:

- **P (Present):** Regular full-day attendance on site.
- **A (Absent):** Unexcused absence or approved unpaid leave.
- **HD (Half Day):** 4 hours duty completed.
- **OD (On Duty):** Approved field operations, flight trials, or client demos.
- **OT (Overtime):** Approved extended hours beyond standard shift.

PHASE 2: FUTURE KEY-CARD INTEGRATION (ROADMAP: 1 MONTH)

Target: Deployment within 1 month.
Architecture: RFID Key-Card / biometric access hardware installed at facility turnstiles. Hardware automatically logs employee tap-in and tap-out timestamps directly into the platform database, auto-calculating P/HD/OT without manual HR intervention.

VERIFY HARDWARE / API COMPATIBILITY BEFORE CODING

MONTHLY SALARY SHEET & PAYROLL COMPILATION ENGINE

At month-end, the system compiles daily attendance records (Total Present days + OD + OT hours - Half-day deductions - LOP). It auto-calculates gross earnings, attendance deductions, and net pay, presenting a preliminary Salary Sheet for Ganesh and Maneesha's review and approval.

SECTION 26 • HR AUTOMATION & PAYROLL CONTROLS

Automated HR Document Generation & Payroll Overrides

Dynamic generation of employment documentation, automated salary deductions, and authorized monthly net pay modifications.

1. Offer Letters — CONFIRMED

Dynamic Assembly: System auto-generates formal employment Offer Letters pulling candidate name, CTC package, joining date, and the specific Job Description configured in the platform profile.

2. Experience Letters — CONFIRMED

Dynamic Assembly: Auto-populates tenure dates, official designations, department achievements, and separation clearance status upon employee request and HR admin approval.

3. Digital Payslips — CONFIRMED

Dynamic Assembly: Generated monthly breaking down basic pay, HRA, special allowances, attendance LOP deductions, OT pay, and final net disbursement amount.

PAYROLL ENGINE & MANDATORY MONTHLY PAY ADJUSTMENT CAPABILITY

CONFIRMED REQUIREMENT

1. Automated Calculation Engine

- Base Salary fetched from Employee Profile.
- Attendance deductions auto-computed from Absent (A) and Half-Day (HD) records.
- Overtime (OT) additions auto-tallied based on approved OT logs.
- Preliminary Monthly Net Pay = $Base - Deductions + OT$.

2. Admin Monthly Pay Override Capability

Confirmed Requirement: Ganesh and Maneesha possess the explicit system authority to manually modify or adjust the final net monthly payout figure (e.g. if a discretionary cash bonus, special advance recovery, or non-standard amount was given instead of the auto-calculated figure).

SELF-SERVICE HR DOCUMENT REQUEST & RELEASE WORKFLOW

1. Employee Portal: Employee clicks "Request Payslip / Experience Letter" → **2. Auto-Routing:** Request logged to Ganesh Pentela & Uppalapati Maneesha → **3. Review / Pay Adjustment:** HR verifies records, applies adjustments if needed → **4. Release:** PDF generated & delivered into Employee's personal portal

CONFIRMED HR ACCESS BOUNDARY:

General managers, sales leads, and production leads have zero access to employee salary figures, payroll sheets, or HR documents. All HR administrative controls, salary sheet adjustments, and document approvals remain strictly confined to Ganesh Pentela and Uppalapati Maneesha.

SECTIONS 27 & 28 • EMPLOYEE PORTAL & USER INTERFACE

Employee Self-Service (ESS) & Dashboard Mockup

Simplified, uncluttered interface designed specifically for technicians and field engineers — zero complex ERP menus.

Daily Work & Checklists
Displays manager-assigned tasks for the day. Click opens the unit's 20-page database checklist with a single tap.

1-Click HR Requests
Direct action buttons for Leave Application, Payslip Request, and Experience Letter — routes directly to Ganesh & Maneesha.

Petty Cash Claims
Field employees snap bill photos and submit reimbursement claims directly into the accounts queue.

GOAG ESS PORTAL • LOGGED IN AS: BOMMALI HIMAGIRI (DRONE TECHNICIAN • GOAG-E014) — SYSTEM TIME: 10:45 AM • ATTENDANCE: PRESENT

Hello, Himagiri!
You have 2 assigned assembly work orders active today.

+ APPLY LEAVE | REQUEST PAYSLIP | EXP. LETTER

TODAY'S ASSIGNED TASKS & WORK ORDERS

GOAG-5X-001 (5X Heavy-Lift Drone) — IN PROGRESS
Assigned by: Kapil Kumar • Stage: ESC Soldering & Avionics Harnessing
Approx Usual Time: 4.5 hrs • Elapsed: 2.1 hrs (Timesheet tracking active)

GOAG-5X-002 (Airframe Sub-assembly) — QUEUED
Carbon arms torque check • Awaiting BOM material release from Stores

ACTIVE 20-PAGE CHECKLIST: GOAG-5X-001 — PAGE 5 OF 20

- ✓ [P4.1] Motor 1 to 4 winding resistance verified <0.05Ω
- ✓ [P4.2] ESC heat-shrink insulation sealed with thermal paste
- ☑ **[P5.1] Solder main XT90 power lead to PDB (Enter solder temp)**
- ☐ [P5.2] Verify 12V telemetry rail output under 2A load

Checklist ID: CL-5X-001-A — COMMIT & NEXT STAGE →

CONFIRMED UI DESIGN MANDATE — ZERO COGNITIVE OVERLOAD:
Workshop drone assemblers and technicians are not corporate desk accountants. The ESS dashboard deliberately hides financial general ledgers, tax schemas, purchasing negotiations, and complex multi-tier menus. Technicians see only their assigned work, their active unit checklist, and one-click HR service requests.

SECTION 29 • EXECUTIVE COMMAND & CONTROL

Management & Director Executive Cockpit

Real-time enterprise command center providing cross-functional financial, manufacturing, sales, aftersales, and HR governance.

GOAG EXECUTIVE COCKPIT • MASTER CONSOLE (GANESH PENTELA • GOAG-A005) — DAY | WEEK | **MONTH** | YEAR

MONEY IN (RECEIPTS)	MONEY OUT (EXPENSES + PO)	OUTSTANDING RECEIVABLES	NET MONTHLY OPERATING CASH
₹ 42,50,000	**₹ 28,10,000**	**₹ 11,40,000**	**+ ₹ 14,40,000**
+18% vs Last Month • 14 Quotes Paid	Vendor POs: ₹22L • Petty: ₹1.1L	5 Client Balances Pending	Healthy Operating Runway

SALES PIPELINE

- **Active Leads:** 28 Inquiries
- **Quotes Issued:** 12 Auto / 4 Custom
- **Confirmed WO:** 6 Aircraft Units
- **Dealers Active:** 4 Regional Hubs

Manager: Vamshi Teja

PRODUCTION WIP

- **Active Work Orders:** 4 Units (5X)
- **Checklist Stage:** 3 Assembly / 1 Wiring
- **Ready for Flight:** 1 Unit (Saida Rao)
- **QC Pending:** 1 Unit (Vishwateja)

Lead: Kapil Kumar

AFTERSALES & REPAIR

- **Open Support Calls:** 7 Cases
- **Resolved over Phone:** 4 Tickets
- **Branch Repairs:** 2 Units in Bench
- **External Sent:** 1 Battery at Vendor

Supervisor: Meegala Vikas

HR & WORKFORCE

- **Total Headcount:** 23 Staff
- **Present Today:** 21 (P) / 1 (OD)
- **On Leave:** 1 (Approved Leave)
- **Pending Approvals:** 2 Payslips

Admin: Ganesh & Maneesha

Marketing & Events: Next Expo Demo: 24-Oct • Dealer Meet: 05-Nov | **R&D Project:** Graydon V2 Carbon Hybrid in Bench Testing

[Director ESS Quick-Link: Apply Leave / Personal Payslips]

CONFIRMED MANAGEMENT DASHBOARD OBJECTIVE:

Designed specifically for technology founders and directors to assess company health in 60 seconds without wading through sub-menus. The screen synthesizes money in/out, assembly bottlenecks, customer grievances, and workforce attendance across daily, weekly, monthly, and annual time slices.

SECTION 30 • EXECUTIVE VISUALIZATION & FLEET FOOTPRINT

Geographic Delivery Map & Pincode Telemetry

Nationwide visualization of delivered aircraft units plotted automatically from dispatch postal PIN codes.

Automated Pincode Extraction — CONFIRMED

When Stores registers a dispatch (Customer Pickup or Logistics Shipping), the system extracts the customer's 6-digit postal PIN code and binds it permanently to the aircraft unit record.

Executive Footprint Tracking — CONFIRMED

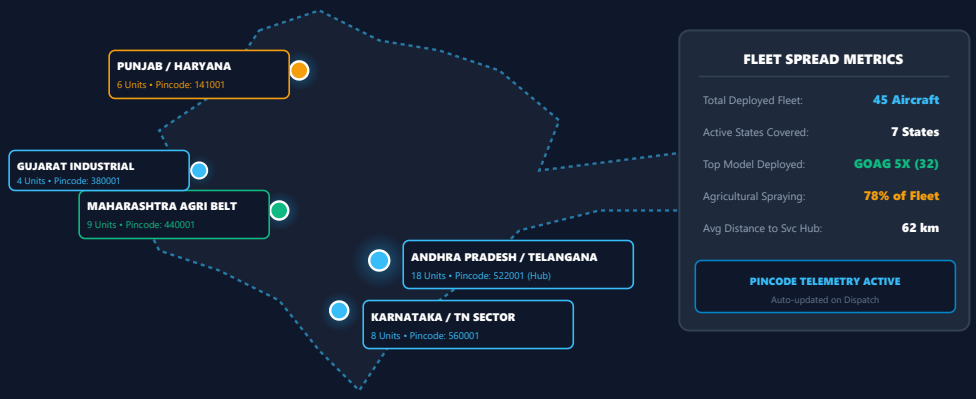
Directors can visualize regional market penetration at a glance. Visual pins display where GOAG 5X and Graydon drones are actively operating across Indian agricultural and industrial belts.

Service Hub Proximity — PROPOSED ENHANCEMENT

Correlates delivery cluster pins against nearest authorized GOAG dealers and service branches to optimize field support response times and mobile spare parts inventory staging.

GOAG COMMERCIAL DRONE DEPLOYMENT MAP (EXECUTIVE VIEW)

FILTER: ALL MODELS • NATIONWIDE DEPLOYMENTS



ARCHITECTURAL CLARIFICATION — MAP ENGINE IMPLEMENTATION:

Per executive design requirements, this map is not an invasive real-time flight telemetry tracking system; it is a clean, executive dashboard feature that visualizes customer delivery locations plotted from dispatch PIN codes. It provides technology leadership with an immediate visual sense of regional fleet growth and operational territory expansion.

SECTION 31 • PARTNER NETWORK & DISTRIBUTION OPERATIONS

Authorized Dealer Portal & Distribution Workflows

Dedicated external portal for ordering aircraft, spare parts catalogs, real-time ledgers, service tickets, and marketing schedules.

1. Drone & Spares Orders

Dealers place bulk orders for complete drones and browse the full spare parts catalog with negotiated distributor discounts.

2. Payment & Running Ledger

Dealers log payment transaction references (UTR); Accounts cross-checks bank credits and updates the running ledger.

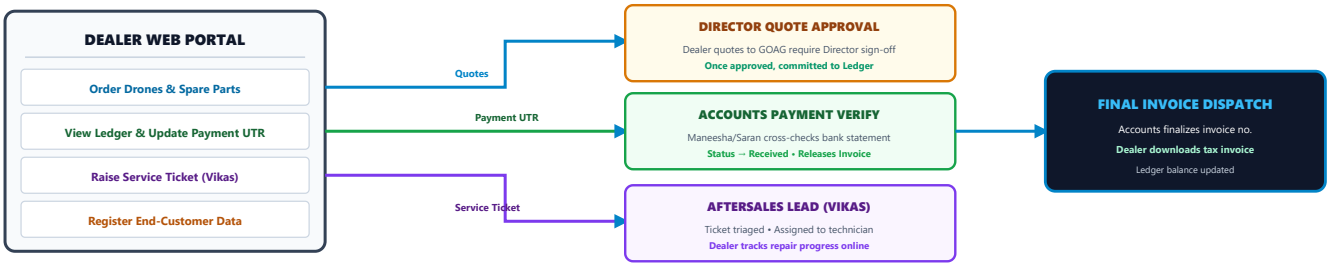
3. Service Tickets to Vikas

Dealers raise customer warranty or repair requests; system routes tickets directly to Aftersales Lead Meegala Vikas.

4. Marketing & Schedules

Access authorized GOAG marketing schedules, promotional collateral, regional demo flight plans, and expo calendars.

DEALER PORTAL FUNCTIONAL INTERFACES & INTERNAL HANDSHAKE ARCHITECTURE



CONFIRMED DEALER QUOTATION APPROVAL RULE:

Dealers can view standard price catalogs for spares and complete aircraft. However, if a dealer submits a custom quotation request or seeks non-standard commercial margins from GOAG, **Director Approval is strictly mandatory** before the quotation is converted into a ledger commitment.

SECTIONS 32 & 33 • DOWNSTREAM TRACEABILITY & CUSTOMER INTERFACE

Dealer Customer Mapping & Customer Portal UI

Binding secondary retail sales to GOAG master records, manufacturer warranty activation, and customer dashboard mockup.

SECTION 32: DEALER CUSTOMER REGISTRATION & TRACEABILITY

CONFIRMED REQUIREMENT

When a dealer sells a drone to an end-customer, the dealer enters the buyer's details through their portal. The system binds this record to GOAG's Master Database:

AUTHORIZED DEALER → END CUSTOMER INFO → DRONE / PRODUCT RECORD → WARRANTY ACTIVATION → AFTERSALES ELIGIBILITY

🔴 **DECISION REQUIRED • SERIAL NUMBER LIFECYCLE:** While the concept of drone unit identifiers exists (e.g. GOAG-5X-001), GOAG must finalize whether a global barcode/QR serial number architecture should track sub-components (motors, batteries) individually.

GOAG CUSTOMER FLEET PORTAL • WELCOME: KRISHNA AGRI ENTERPRISES (CUST-8492)

ACTIVE DRONES: 2 • WARRANTY: VALID

REGISTERED AIRCRAFT: GOAG 5X HEAVY-LIFT (SN: 5X-2026-001)

- **Purchase Date:** 12-Jul-2026 • **Delivered By:** Logistics Shipping
- **Configuration:** 16L Spray Rig + Dual 22000mAh Battery System
- **Total Flight Hours:** 42.5 hrs logged • **Next Scheduled Check:** 50 hrs

OFFICIAL COMPLIANCE & TAX VAULT

📜 **WARRANTY CERTIFICATE**
Uploaded by GOAG Admin [PDF]

📄 **TAX INVOICE & PI**
GST Paid Invoice [Download]

LIVE SERVICE TRACKER: CASE #SRV-2026-0842

✓ Intake & OTP ✓ Diagnostics • In Service (Bench) • Ready • Handover

Technician: Himagiri • Replaced Motor 2 Bearing • Bench Testing Active

SCHEDULED MAINTENANCE REMINDERS

- **50-Hour Motor Inspection:** Due in 7.5 flight hours.
- **Nozzle Cleaning:** Completed on 18-Sep-2026.
- **Battery Cycle Calibration:** Recommended every 25 charge cycles.

SECTION 34 • ENGINEERING INNOVATION & PROTOTYPING

Research & Development (R&D) Workspace

Flexible, unstructured experimentation sandbox completely isolated from rigid commercial production constraints.

Strict Production Isolation — CONFIRMED

Core Rule: R&D cannot be forced into standard Sales → Work Order → Delivery paths. Prototyping requires agile experimentation without commercial invoicing or fixed BOM lockdowns.

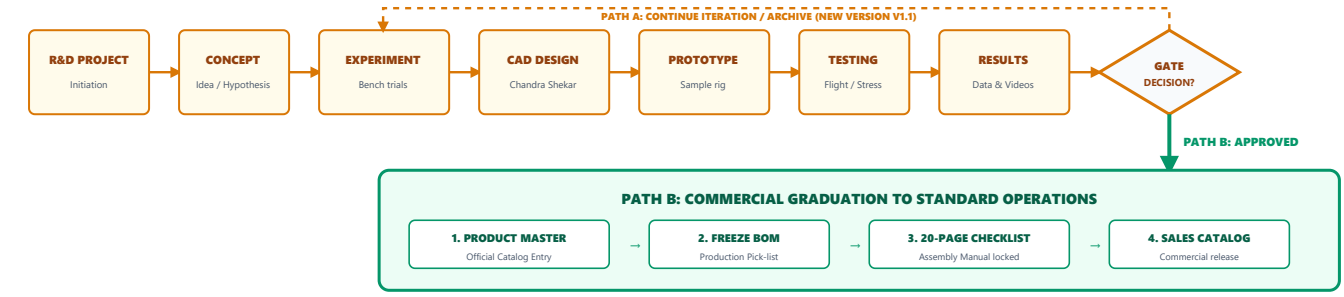
R&D Leadership & Scope — CONFIRMED

Lead: Voleti Veera Babu (GOAG-E009) and Darla Chandra Shekar (Design Engineer). Supports CAD drawings, telemetry logs, test videos, sensor benchmarking, and experimental component costing.

Bifurcated Decision Paths — CONFIRMED

Upon prototype testing, engineering reaches a formal evaluation gate: **Path A** (Continue iterative R&D) or **Path B** (Graduation into commercial Product Master).

R&D AGILE EXPERIMENTATION LOOP → COMMERCIAL GRADUATION GATE



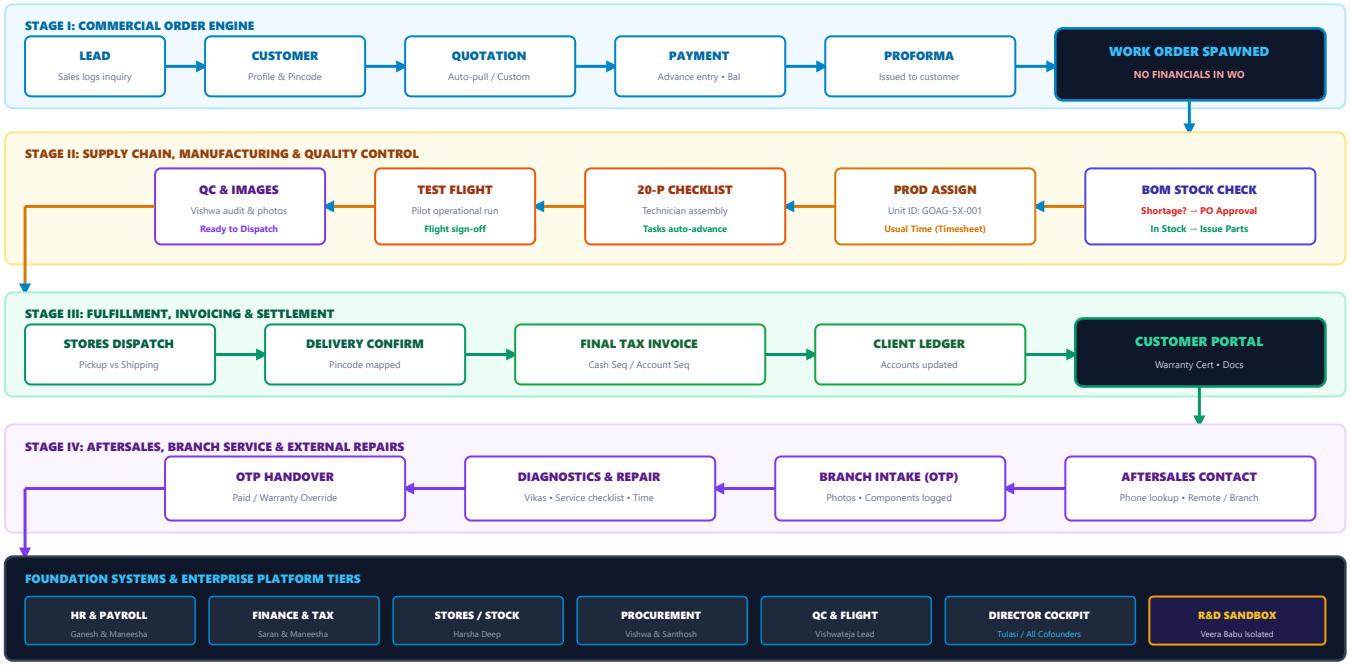
DECISION REQUIRED — FORMAL R&D GRADUATION GATEWAYS:

GOAG must establish: (1) Who officially signs off on moving a prototype from R&D into the commercial Product Master (Director vote vs Technical Lead sign-off)? (2) What minimum flight-test hours are required before freezing a new model's Bill of Materials? (3) How are R&D component consumption costs accounted for in the monthly financial ledger?

SECTION 35 • THE ENTERPRISE OPERATING SYSTEM

The Complete End-to-End GOAG Master Map

The definitive "One-Page Picture" synthesizing every commercial, manufacturing, fulfillment, aftersales, and foundational system.



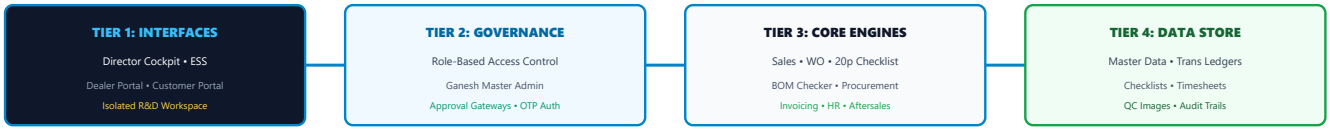
SECTIONS 36 & 37 • PLATFORM TOPOLOGY & RELATIONAL SCHEMA

Information Architecture & Record Relationships

Tiered software layers, secure data persistence boundaries, and relational entity mappings from Customer to Supply Chain.

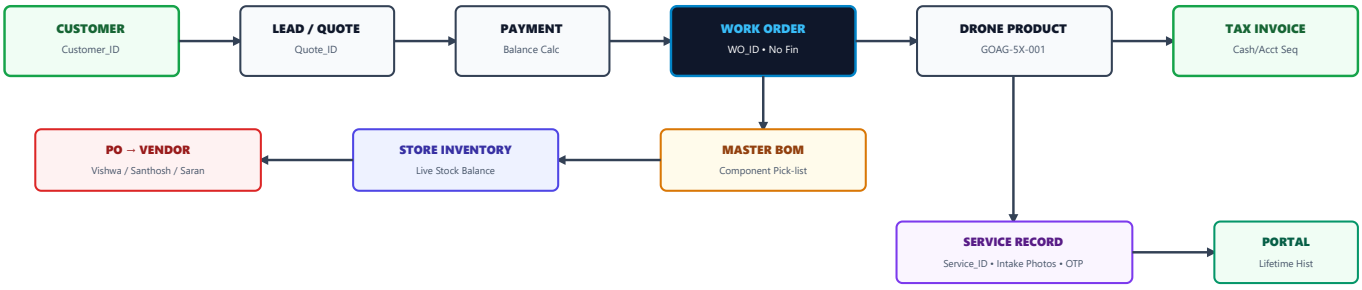
SECTION 36: 4-TIER INFORMATION ARCHITECTURE

CONFIRMED REQUIREMENT



SECTION 37: CORE RECORD RELATIONSHIPS & SYSTEM DATA FLOWS

CONFIRMED REQUIREMENT



SECTIONS 38 & 39 • SYSTEM ENTITIES & STATE MACHINES

Document Numbering & Status Lifecycles

Exhaustive inventory of confirmed entity identifiers, open syntax decisions, and proposed lifecycle state machines.

SECTION 38: ENTITY IDENTIFIER INVENTORY & SOURCE STATUS

CONFIRMED VS DECISION

Entity Identifier	Confirmed Source Syntax	Operational Governance	Status Classification
Employee Codes	`GOAG-A001` to `GOAG-A006` `GOAG-E007` to `GOAG-E023`	Universal primary key for all 23 personnel; binds RBAC, timesheets, and HR sheets.	CONFIRMED
Product Work Identifier	Concept: `GOAG-5X-001`	Unique hardware unit code during 20-page assembly and flight testing.	CONFIRMED CONCEPT
Cash Invoices	Dedicated Cash Sequence	Sequential number series exclusively for cash receipts; credits physical cash book.	CONFIRMED SPLIT
Account (Bank) Invoices	Dedicated Account Sequence	Sequential number series exclusively for bank/NEFT/RTGS; credits bank registers.	CONFIRMED SPLIT
Customer ID	`[Format Undefined]`	Unique client identifier linking leads, orders, warranties, and portal accounts.	DECISION REQUIRED
Work Order ID	`[Format Undefined]`	Production floor build traveler linking BOM items and assembly checklists.	DECISION REQUIRED
Service ID	`[Format Undefined]`	Branch service ticket linking intake condition photos, technician remarks, and OTPs.	DECISION REQUIRED

SECTION 39: PROPOSED SYSTEM STATUS STATE MACHINES

PROPOSED STATUS MODEL — TEAM TO VERIFY

1. Drone Manufacturing & Fulfillment State Machine

WO Created → BOM Checked → Assigned → In Assembly (20p) → Ready Test Fly → Ready QC → Ready Dispatch → Delivered

2. Component Procurement State Machine

MR Logged → Vishwa Review → Director Approved → PO Issued → Inward QC Passed → Stocked

3. Branch Aftersales & Service State Machine

Intake Photos → Customer OTP → Diagnostics (Vikas) → In Repair (Checklist) → Repair Tested → Handover OTP

🔴 **DECISION REQUIRED — EXACT SYNTAX STANDARDIZATION:**
Prior to engineering database tables, the GOAG Board must approve the exact string templates for system IDs (e.g. `GOAG-WO-YYYY-XXXX` for Work Orders, `GOAG-SRV-YYYY-XXXX` for Service cases, and `CUST-XXXXXX` for customer profiles).

SECTION 40 • PROCESS ENGINEERING & OPERATIONAL GOVERNANCE

Automation vs. Manual Responsibility Matrix

Exhaustive classification of system processes into manual human actions, automated software triggers, and mandatory approvals.

Operational Process	Manual Human Action	Automated Software Action	Approval Authority	Source Basis
Lead & Quote Intake	Sales inputs customer requirements; enters custom pricing if needed.	Pulls catalog prices, HSN, and calculates dynamic GST (inter/intrastate).	Director (if custom price)	CONFIRMED
Payment & Proforma	Sales logs advance payment received from customer.	Auto-calculates remaining balance; generates Proforma Invoice (PI).	None (System Auto)	CONFIRMED
Work Order Spawning	Sales sets Commercial Estimated Delivery Date (EDD).	Spawns WO with unique ID; strips financial figures for technicians.	None (System Auto)	CONFIRMED
BOM Stock Audit	None (fully autonomous check triggered on WO creation).	Parses multi-level BOM against physical stock; flags shortages to Stores.	None (System Auto)	CONFIRMED
Component Purchase	Vishwateja selects supplier; enters remarks if alternate vendor chosen.	Generates PO; auto-tags to Maneesha & Saran; commits to monthly expense.	Vishwa / Santhosh / Saran	CONFIRMED
Assembly Checklist	Technician verifies assembly steps across 20-page checklist.	Logs completion timestamps; loads next stage into technician's pending work.	Production Manager	CONFIRMED
QC & Photo Audit	Inspector conducts physical audit; captures and uploads multi-angle photos.	Enforces image attachment before permitting "Ready to Dispatch" state.	Head QC (Vishwateja)	CONFIRMED
Stores Dispatch	Stores packages drone; verifies customer pickup vs courier shipping.	Extracts destination postal PIN code; maps pin on Director Delivery Map.	None (Stores Action)	CONFIRMED
Final Tax Invoicing	Maneesha validates all customer data, E-Way Bill, and tax ledger.	Allocates next sequential number (Cash Seq vs Account Seq); posts ledger.	Maneesha / Saran / Ganesh	CONFIRMED
Delayed Payment	Sales requests commercial credit for institutional client.	Holds final dispatch release until management authorization token committed.	Core Management Team	CONFIRMED
Customer Portal Sync	None (system provisions account upon first order).	Auto-syncs order progress, invoices, and uploads official warranty certs.	None (System Auto)	CONFIRMED
Aftersales Phone Call	Technician enters caller phone; logs defect description and advice.	Auto-pops customer fleet profile, purchase date, and warranty status.	None (Support Action)	CONFIRMED
Branch Inward (OTP)	Stores snaps condition photos; catalogs accessories brought along.	Dispatches intake OTP to customer mobile; binds case to customer portal.	None (Customer OTP)	CONFIRMED
Service Diagnostics	Vikas assigns technician; technician logs diagnostics & estimated cost.	Records approximate usual service time to monitor employee timesheets.	Aftersales Lead (Vikas)	CONFIRMED
Service Override	Vikas applies warranty waiver or commercial discount with remarks.	Bypasses billing requirement; unlocks customer handover OTP flow.	Aftersales Lead (Vikas)	CONFIRMED
External Repairs	Vikas routes batteries/chargers to vendor; logs date shipped & tracking.	Maintains external custody loop; sends progress updates to customer portal.	Aftersales Lead (Vikas)	CONFIRMED
Daily Attendance	Maneesha/Ganesh logs daily status (P, A, HD, OD, OT) manually.	Compiles total days worked; auto-calculates monthly loss-of-pay deductions.	Ganesh & Maneesha	CONFIRMED (RFID Future)
Payroll Adjustment	Ganesh/Maneesha modifies monthly pay if custom bonus/amount awarded.	Generates dynamic payslips, salary sheets, and bank payout registers.	Ganesh & Maneesha	CONFIRMED
Dealer Quotes	Dealer submits custom pricing or bulk discount request.	Locks quotation until authorized; commits to dealer ledger upon approval.	Directors Approval	CONFIRMED

SECTIONS 41 & 42 • GOVERNANCE GATEWAYS & EVENT MESSAGING

Approval Hierarchy & Notification Architecture

Formal managerial sign-off decision trees and segregation of confirmed transaction alerts from proposed notification engines.

SECTION 41: AUTHORITATIVE MANAGERIAL APPROVAL GATES

CONFIRMED GOVERNANCE

1. Custom Quotations
Approver: Tulasi Ram / Saran / Vishwateja.
Trigger: Manual line pricing or discounts deviating from Master Catalog.

2. Delayed Payments
Approver: Core Management (Directors).
Trigger: Client requests delivery with outstanding balance on credit terms.

3. Purchase Orders (PO)
Approver: Vishwa / Santhosh / Saran.
Trigger: Stores material shortage; requires remarks if alternate vendor chosen.

4. Dealer Special Terms
Approver: Executive Directors.
Trigger: Dealer submits custom quotation or non-standard margin requests.

5. Service Warranty Waivers
Approver: Meegala Vikas (Lead Aftersales).
Trigger: Repair fee waived under warranty or approved commercial discount.

6. HR Leaves & Documents
Approver: Ganesh Pentela & Maneesha.
Trigger: Leave applications, payslip modifications, and experience letters.

SECTION 42: NOTIFICATION ARCHITECTURE (CONFIRMED VS. PROPOSED)

CONFIRMED | PROPOSED ENGINE

CONFIRMED NOTIFICATION TRIGGERS

- **Quote Sharing:** Automatic email and SMS text with secure link to view/download quotation PDF.
- **Portal Onboarding:** Welcome credentials sent to new customer upon order confirmation.
- **Intake & Handover OTP:** Real-time SMS OTP sent to customer mobile for branch service.
- **PO Accounts Tagging:** Instant system alert notifying Maneesha & Saran of committed PO liabilities.
- **HR Routing:** Employee leave and document requests immediately notify Ganesh & Maneesha.

PROPOSED NOTIFICATION ENGINE — NOT YET APPROVED

- **WhatsApp Business API:** Automated delivery of invoice PDFs and order milestones via WhatsApp.
- **Low-Stock Reorder Triggers:** Automated push notifications when fast-moving spares breach safety stock.
- **Preventive Maintenance Alerts:** Automated SMS alerts notifying operators when flight hours reach service intervals.
- **In-App Activity Bell:** Browser push notifications for task assignments on technician dashboards.

STRICT COMPLIANCE RULE:
Only confirmed notifications (Email/SMS for quotes, portal logins, and OTPs) will be built in Phase 1. The proposed multi-channel notification engine (WhatsApp API, push alerts) remains pending formal technical approval by the GOAG Board.

SECTION 43 • DATA GOVERNANCE & HISTORICAL IMMUTABILITY

Audit Trail & Historical Record Logging

Exhaustive inventory of mandated historical logs and enterprise specification for a proposed immutable audit engine.

CONFIRMED HISTORICAL LOGGING OBLIGATIONS

CONFIRMED REQUIREMENT

Historical Log Domain	Logged Data Entities & Artifacts	Operational & Analytical Purpose
Aftersales Support & Defects	Every customer call, symptom type, advice given, phone resolution vs branch visit.	**Pattern Analysis:** Detects recurring hardware/firmware failures to feed R&D.
Petty Cash & Bills Vault	Scanned receipts, vouchers, employee reimbursement claims, approving accountant ID.	**Financial Compliance:** Prevents unreceipted cash leakage; audit-ready expense books.
Customer & Dealer Ledgers	Advance receipts, bank UTRs, invoice debits, payment credits, running balances.	**Accounts Settlement:** Permanent financial single source of truth across transactions.
Purchase Orders & Approvals	Approver identity (Vishwa/Santhosh/Saran), supplier chosen, alternate vendor remarks.	**Procurement Governance:** Complete traceability on vendor price selections.
Work Order Build History	Assigned technician, step-by-step 20-page checklist timestamps, flight test logs, QC photos.	**Aerospace Traceability:** Complete unit pedigree from bare carbon tubes to certified aircraft.
Timesheet Work Durations	Approximate usual time assigned vs actual technician elapsed build duration.	**Workforce Discipline:** Understands employee work patterns; internal monitoring.

PROPOSED ADDITION — NOT YET APPROVED: CENTRALIZED IMMUTABLE AUDIT TRAIL

PROPOSED ADDITION

Architectural Capability (Proposed)

An automated, system-wide Change Data Capture (CDC) engine that creates an append-only, immutable audit log for every critical table in the database.

• Captures: `Timestamp, User_ID, IP_Address, Action (Create/Update/Delete), Previous_Value, New_Value`.

• Prevents silent administrative alteration of Master Data, pricing, or tax slabs without permanent audit logging.

Target Entities for Complete Audit Logging

- **Master Data Tables:** Base prices, component costs, HSN codes, GST percentages.
- **Financial Ledgers:** Cash vs Account invoice records, manual petty cash adjustments.
- **Work Order Traveler:** Technician checklist sign-offs, QC approval timestamps.
- **HR & Payroll:** Attendance status changes, manual monthly net pay adjustments.

AUDIT COMPLIANCE STATEMENT:

All functional historical records (Service patterns, petty cash bills, customer ledgers, and build checklists) are strictly confirmed requirements that will be fully implemented. The proposed system-wide CDC engine is presented for board consideration to enhance regulatory compliance.

SECTION 44 • STRATEGIC GAPS & AMBIGUITIES

Decisions Required Before Software Development

The consolidated registry of operational ambiguities and policy choices that GOAG leadership must formally decide.

1. Final Invoice vs. Proforma Trigger

HIGH PRIORITY

Resolve the source conflict: Does delivery trigger an auto-PI if zero balance, or is a final GST Tax Invoice issued upon physical dispatch for all deliveries?

2. System ID & Numbering Standards

ARCHITECTURE

Approve final string templates for `Customer_ID`, `WO_ID`, `Service_ID`, and Cash vs Account invoice prefix conventions.

3. Statutory Indian GST Rule Engine

STATUTORY / TAX

Engage company CA to formally audit place-of-supply rules, HSN codes for drone frames vs agricultural sprayer kits, and E-Way Bill thresholds.

4. Drone Hardware Serial Numbering

MANUFACTURING

Decide whether serialized barcode tracking is enforced only for completed airframes (e.g. GOAG-5X-001) or extended to individual motors, ESCs, and batteries.

5. QC Rejection & Rework Routing

QUALITY

Define formal routing when QC fails: Does the drone return to the original assembler? Does it require a complete 20-page re-inspection or modular fix?

6. Support Staff Rotation Model

OPERATIONS

Define eligible employee pool for previous-day aftersales rotation: Are workshop assembly technicians included, or only commercial support personnel?

7. R&D Commercial Graduation Gateways

R&D / GOVERNANCE

Establish formal sign-off protocol for transitioning an experimental prototype into commercial Product Master and freezing its official Bill of Materials.

8. Dealer Payment Reconciliation SLA

FINANCE

Specify internal SLA for Accounts (Maneesha/Saran) to cross-check dealer-entered UTR payment claims against bank statements before order release.

9. Key-Card Hardware Specifications

HARDWARE ROADMAP

Select RFID / biometric hardware vendor and confirm API communication protocol (HTTP webhook vs local TCP socket) for 1-month attendance deployment.

10. Media Storage & Cloud Retention

INFRASTRUCTURE

Establish retention policies and storage architecture (AWS S3 / Wasabi / Cloudflare R2) for high-res QC photos and branch intake damage evidence.

MANDATORY PRE-DEVELOPMENT FREEZE RULE:
None of these 10 decisions should be resolved by software developers during coding. The GOAG Board and operational leadership must formalize and sign off on each item during the draft review meeting to ensure the software matches exact business expectations.

SECTION 45 • ARCHITECTURAL ENHANCEMENTS (PENDING APPROVAL)

Proposed Architectural Additions & System Upgrades

High-value enterprise enhancements recommended for consideration — strictly labeled as proposed and not yet approved.

ARCHITECTURAL INTEGRITY NOTICE: None of the features below are present in the core source requirements. They are proposed modern engineering enhancements designed to scale GOAG's operations. They must be explicitly approved or rejected during board review.

1. Unified Global Search Bar

PROPOSED ADDITION

A header-mounted universal search indexing Customer Names, Phone Numbers, Work Orders, Drone Serials, and Invoices with instant sub-second lookup.

2. WhatsApp Business API Messaging

PROPOSED ADDITION

Automated multi-channel messaging delivering Quotation links, Tax Invoices, and OTP verifications directly into client and dealer WhatsApp chats.

3. Immutable Centralized Audit Trail

PROPOSED ADDITION

Change Data Capture (CDC) logging recording every database update with user, IP, timestamp, old value, and new value for regulatory audit readiness.

4. Headless PDF Document Engine

PROPOSED ADDITION

Server-side automated PDF generation producing pixel-perfect, watermarked Quotations, Tax Invoices, E-Way bills, Payslips, and Warranty Certificates.

5. Dynamic Multi-Level Approval Engine

PROPOSED ADDITION

Configurable approval matrices allowing directors to adjust financial authorization thresholds (e.g. POs >₹1,00,000 require two director approvals).

6. Entity Activity Timelines

PROPOSED ADDITION

Chronological visual audit feed displayed on customer profiles and Work Orders showing who called, who edited specs, and who signed off checklists.

7. Centralized Media Storage Vault

PROPOSED ADDITION

Secure S3-compatible cloud object store with automatic image compression and thumbnail generation for QC inspection photos and branch damage receipts.

8. Automated Backup & Disaster Recovery

PROPOSED ADDITION

Automated hourly database snapshots, point-in-time recovery (PITR), and encrypted off-site replication to ensure zero business disruption.

9. Hardware Version Control for R&D

PROPOSED ADDITION

Semantic versioning (v1.0, v1.1) and Git-backed repository management for CAD files, PCB schematics, and flight controller parameter sets.

10. Formal Return & Replacement (RMA)

PROPOSED ADDITION

Structured Return Merchandise Authorization module for tracking dead-on-arrival components, warranty core exchanges, and reverse logistics.

SECTION 46 • FORMAL OPERATIONAL APPROVAL GATE

Team Review Checklist & Executive Sign-Off

Exhaustive verification checklist across all functional modules and formal executive sign-off before software engineering begins.

BEFORE WE START DEVELOPMENT — OPERATIONAL VERIFICATION CHECKLIST

27-POINT AUDIT

- [x] Sales flow confirmed
- [x] Payment flow confirmed
- [x] Work Order flow confirmed
- [x] Production flow confirmed
- [x] Checklist flow confirmed
- [x] Test flight flow confirmed
- [x] QC flow confirmed
- [x] Stores flow confirmed
- [x] Inventory flow confirmed
- [x] Purchase flow confirmed
- [x] Accounts flow confirmed
- [x] Invoice flow confirmed
- [x] Aftersales flow confirmed
- [x] External service flow confirmed
- [x] HR flow confirmed
- [x] Employee dashboard confirmed
- [x] Management dashboard confirmed
- [x] Dealer portal confirmed
- [x] Customer portal confirmed
- [x] R&D workflow confirmed
- [x] Employee permissions confirmed
- [x] Approval hierarchy confirmed
- [x] IDs & Numbering confirmed
- [x] GST/tax implementation verified
- [x] Dashboard KPIs confirmed
- [x] Missing requirements identified
- [x] Proposed additions reviewed

EXECUTIVE BOARD APPROVAL & FREEZE SIGN-OFF

By signing below, the executive directors confirm that this document accurately represents how GOAG Services intends to operate across all departments.

Employee Code	Director Name	Executive Designation	Signature	Date
GOAG-A001	**Yalamarthi Tulasi Ram**	Cofounder, Director		___/___/2026
GOAG-A002	**Bandaru Santhosh Kumar**	Cofounder, Director		___/___/2026
GOAG-A003	**V Vishwateja Kumar Reddy**	Cofounder, Head Quality Control		___/___/2026
GOAG-A004	**Battula Saran Kumar**	Cofounder, Head of Finance		___/___/2026
GOAG-A005	**Ganesh Pentela**	Cofounder, Head - HR & Operations (Master Admin)		___/___/2026
GOAG-A006	**Vamsi Krishna K**	Cofounder, Head Aftersales		___/___/2026

FINAL TEAM ALIGNMENT QUESTION:

“IS THIS EXACTLY HOW WE WANT GOAG TO OPERATE BEFORE WE START BUILDING THE SOFTWARE?”

If YES: Lock this architecture blueprint and initiate software development. If NO: Amend the specific workflow section prior to coding.